

**NOV/DEC 2026 EXAMINATION REGULAR**  
**B.E / B.TECH**  
**COMMON TO CSE, CSE(AIML),IT,AIDS**  
**5<sup>TH</sup> Semester**  
**R2023\_V 1.1**

| COURSE CODE | COURSE TITLE     |
|-------------|------------------|
| 2311CSE906T | UI AND UX DESIGN |

CO1: Apply UI and UX design principles to develop user-centered application designs  
CO2: Design visually consistent and accessible user interfaces using UI design principles  
CO3: Analyze UX processes, user needs, and research methods to improve product user experience.  
CO4: Construct wireframes and interactive prototypes using sketching, responsive design, and usability testing techniques for prototyping  
CO5: Develop UX solutions using research methods, information architecture, and prototype evaluation techniques  
K1 – Remember    K2 – Understand    K3 – Apply    K4 – Analyse    K5 – Evaluate    K6 – Create

**UNIT- 1**  
**FOUNDATIONS OF DESIGN**  
**Part A**

| Q.No  | Questions                                                                                                                                                                                                                                                                         | CO's | Unit | Bloom's Level |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1.    | <b>What is the primary focus of UI (User Interface) design?</b>                                                                                                                                                                                                                   | CO1  | 1A   | K2            |
| 1 Ans | The primary focus of UI design is to enhance the visual appeal and interactivity of a product. It involves designing the elements that users interact with directly, such as buttons, icons, and other graphical elements.                                                        | CO1  | 1A   | K2            |
| 2.    | <b>How does UX (User Experience) design differ from UI design?</b>                                                                                                                                                                                                                | CO1  | 1A   | K2            |
| 2 Ans | UX design is a broader concept that encompasses the overall experience a user has with a product. It includes UI design but goes beyond, considering factors like user research, information architecture, and the overall journey a user takes while interacting with a product. | CO1  | 1A   | K2            |
| 3.    | <b>Give an example of a UI element.</b>                                                                                                                                                                                                                                           | CO1  | 1A   | K1            |
| 3 Ans | Buttons, sliders, and navigation menus are examples of UI elements. These are visual elements that users interact with to perform specific actions or navigate through a product.                                                                                                 | CO1  | 1A   | K1            |
| 4.    | <b>What role does empathy play in UX design?</b>                                                                                                                                                                                                                                  | CO1  | 1A   | K2            |
| 4 Ans | Empathy in UX design involves understanding and sharing the feelings of users. Designers who empathize with users can create products that address their needs and pain points, ultimately leading to a more user-centric and successful design.                                  | CO1  | 1A   | K2            |
| 5.    | <b>Give an example of a UX research method</b>                                                                                                                                                                                                                                    | CO1  | 1A   | K2            |
| 5 Ans | User interviews, usability testing, and surveys are examples of UX research methods. These techniques help designers gather valuable insights into user behavior, preferences, and                                                                                                | CO1  | 1A   | K2            |

|               |                                                                                                                                                                                                                                                                                           |            |           |           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|               | expectations.                                                                                                                                                                                                                                                                             |            |           |           |
| <b>6.</b>     | <b>Name the first stage of Design Thinking.</b>                                                                                                                                                                                                                                           | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>6 Ans</b>  | The first stage of Design Thinking is the "Empathize" stage. It involves understanding the user's needs and gaining insights into their experiences.                                                                                                                                      | CO1        | 1A        | K1        |
| <b>7.</b>     | <b>What is the purpose of the "Define" stage in Design Thinking?</b>                                                                                                                                                                                                                      | <b>CO1</b> | <b>1A</b> | <b>K2</b> |
| <b>7 Ans</b>  | The "Define" stage is focused on synthesizing the information gathered during the Empathize stage to define the core problems and challenges that need to be addressed in the design process.                                                                                             | CO1        | 1A        | K2        |
| <b>8.</b>     | <b>In which stage of Design Thinking do designers ideate and brainstorm solutions?</b>                                                                                                                                                                                                    | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>8 Ans</b>  | Ideation and brainstorming occur in the "Ideate" stage of Design Thinking. This stage encourages creative thinking to generate a wide range of potential solutions.                                                                                                                       | CO1        | 1A        | K1        |
| <b>9.</b>     | <b>What role does prototyping play in the Design Thinking process?</b>                                                                                                                                                                                                                    | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>9 Ans</b>  | Prototyping is a key activity in the "Prototype" stage of Design Thinking. It involves creating tangible representations of ideas to quickly test and iterate on potential solutions.                                                                                                     | CO1        | 1A        | K1        |
| <b>10.</b>    | <b>Which stage focuses on testing and refining solutions with end-users?</b>                                                                                                                                                                                                              | <b>CO1</b> | <b>1A</b> | <b>K2</b> |
| <b>10 Ans</b> | The "Test" stage is dedicated to testing and gathering feedback on prototypes from end users. This iterative process helps refine and improve the design based on real user experiences.                                                                                                  | CO1        | 1A        | K2        |
| <b>11.</b>    | <b>Differentiate Divergent and Convergent Thinking?</b>                                                                                                                                                                                                                                   | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>11 Ans</b> | Divergent thinking is about generating a multitude of ideas, fostering creativity, and exploring possibilities, while convergent thinking involves analyzing and selecting the best solution from those generated, emphasizing logical and systematic decision making.                    | CO1        | 1A        | K1        |
| <b>12.</b>    | <b>Differentiate brainstorming and game storming?</b>                                                                                                                                                                                                                                     | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>12 Ans</b> | While both brainstorming and gamestorming involve group ideation, brainstorming is a more general and open-ended approach, whereas gamestorming introduces structure and interactive elements through specific activities or games to enhance creativity and engagement.                  | CO1        | 1A        | K1        |
| <b>13.</b>    | <b>State the purpose of User Journey Mapping.</b>                                                                                                                                                                                                                                         | <b>CO1</b> | <b>1A</b> | <b>K1</b> |
| <b>13 Ans</b> | The purpose of User Journey Mapping is to visualize the complete experience of a user while interacting with a product or service. It helps identify user needs, pain points, emotions, and opportunities for improvement, enabling organizations to enhance the overall user experience. | CO1        | 1A        | K1        |
| <b>14.</b>    | <b>Mention two types of personas.</b>                                                                                                                                                                                                                                                     | <b>CO1</b> | <b>1A</b> | <b>K2</b> |

|               |                                                                                                                                                                                                                                                                                           |            |           |           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
| <b>14 Ans</b> | Two common types of personas are:<br>Primary Persona – Represents the main target user whose needs are prioritized during the design process.<br>Secondary Persona – Represents additional users who have different needs but are also considered while designing the product or service. | CO1        | 1A        | K2        |
| <b>15.</b>    | <b>List two tools used for Problem Framing.</b>                                                                                                                                                                                                                                           | <b>CO1</b> | <b>1A</b> | <b>K2</b> |
| <b>15 Ans</b> | Two commonly used tools for Problem Framing are:<br>5 Whys Analysis – A technique used to identify the root cause of a problem by repeatedly asking "Why?"<br>How Might We (HMW) Questions – A method used to reframe problems into opportunities for generating innovative solutions.    | CO1        | 1A        | K2        |

#### PART- B

| Q.No         | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO's       | Unit      | Bloom's Level |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|---------------|
| <b>1.</b>    | <b>Discuss the relationship between UI and UX design. How do the core stages of Design Thinking contribute to achieving a harmonious balance between user interface and user experience? Illustrate with examples.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CO1</b> | <b>1B</b> | <b>K3</b>     |
| <b>1 Ans</b> | <b>1. Introduction</b> <ul style="list-style-type: none"> <li><b>UI (User Interface):</b> Visual design – buttons, colors, typography, icons, layout.</li> <li><b>UX (User Experience):</b> User journey – usability, accessibility, efficiency, satisfaction.</li> </ul> <b>2. UI vs UX (3 Marks)</b> <ul style="list-style-type: none"> <li><b>UI:</b> Appearance &amp; interaction.</li> <li><b>UX:</b> Overall user experience.</li> <li>Good UI → <b>Better UX.</b></li> <li>UX research guides UI.</li> <li><b>Success = Good UI + Good UX.</b></li> <li><b>Example:</b> Attractive app + confusing checkout = Poor UX; Clean interface + easy navigation + fast checkout = Good UI &amp; UX.</li> </ul> <b>3. Design Thinking Stages (7 Marks)</b> <ul style="list-style-type: none"> <li><b>Empathize:</b> Understand users (interviews, surveys, observation). → Identify needs; accessible UI.</li> <li><b>Define:</b> Identify problem. → Simplify layout; reduce steps.</li> <li><b>Ideate:</b> Generate solutions. → Better user flow; new layouts (e.g., one-click checkout).</li> <li><b>Prototype:</b> Wireframes/mockups. → Test usability; refine colors, icons, spacing.</li> <li><b>Test:</b> User feedback. → Improve satisfaction; adjust buttons, fonts, consistency.</li> </ul> <b>4. Example (2 Marks) – Food Delivery App</b> <ul style="list-style-type: none"> <li><b>UI:</b> Attractive colors, clear icons, readable text.</li> <li><b>UX:</b> Easy search, quick ordering, smooth payment.</li> <li><b>Design Thinking:</b> Continuous feedback &amp; improvement.</li> </ul> <b>5. Conclusion (1 Mark)</b> <ul style="list-style-type: none"> <li><b>UI = Visual Design</b></li> <li><b>UX = User Satisfaction</b></li> </ul> | <b>CO1</b> | <b>1B</b> | <b>K3</b>     |

|                                    | <ul style="list-style-type: none"><li>• <b>Design Thinking:</b> Empathize → Define → Ideate → Prototype → Test → Better UI &amp; UX.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------|---------------------------|--------------------------------------|-----------------------------|-------------------------------|------------------------------|-----------------------------------------|------------------------------|------------------------------------|------------------------------------|------------------------------------------|-----|----|----|
| 2.                                 | <b>Explain the core stages of Design Thinking and their significance in the design process. How do these stages foster both divergent and convergent thinking, and why is it crucial to incorporate both types of thinking in the design journey?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO1           | 1B           | K3                        |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| 2<br>Ans                           | <b>1. Introduction (2 Marks)</b> <ul style="list-style-type: none"><li>• Design Thinking: Human-centered, iterative problem-solving.</li><li>• Focus: Understand users → Generate ideas → Prototype → Test → Improve.</li></ul> <b>2. Core Stages (5 Marks)</b> <ul style="list-style-type: none"><li>• Empathize: User needs (observation, interviews, research).</li><li>• Define: Identify &amp; frame the problem.</li><li>• Ideate: Brainstorm creative solutions.</li><li>• Prototype: Create low-cost models/mockups.</li><li>• Test: User feedback &amp; refinement.</li></ul> <b>3. Divergent vs Convergent Thinking (3 Marks)</b> <ul style="list-style-type: none"><li>• Divergent Thinking: Many ideas, creativity, exploration. <i>(Empathize, Ideate)</i></li><li>• Convergent Thinking: Analyze, select, refine best solution. <i>(Define, Prototype, Test)</i></li></ul> <b>4. Importance (2 Marks)</b> <ul style="list-style-type: none"><li>• Divergent: Innovation, multiple solutions.</li><li>• Convergent: Practical, feasible, user-focused.</li><li>• Both: Innovative + effective designs.</li></ul> <b>5. Conclusion (1 Mark)</b> <ul style="list-style-type: none"><li>• Design Thinking = Empathize → Define → Ideate → Prototype → Test</li><li>• Balance creativity + evaluation = Practical, user-centered solutions.</li></ul> | CO1           | 1B           | K3                        |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| 3.                                 | <b>Compare and contrast brainstorming and gamestorming as ideation techniques. Provide examples of scenarios where each method might be more suitable in the context of UI/UX design, and explain the benefits of integrating these techniques into the design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CO1           | 1B           | K4                        |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| 3<br>Ans                           | <b>1. Introduction (2 Marks)</b> <ul style="list-style-type: none"><li>• <b>Brainstorming:</b> A technique where participants freely generate as many ideas as possible without criticism.</li><li>• <b>Gamestorming:</b> A structured ideation technique that uses games and interactive activities to encourage creativity and collaboration.</li></ul> <b>2. Comparison between Brainstorming and Gamestorming (4 Marks)</b> <table><tr><th>Brainstorming</th><th>Gamestorming</th></tr><tr><td>Free-flow idea generation</td><td>Uses structured games and activities</td></tr><tr><td>Simple and quick to conduct</td><td>More interactive and engaging</td></tr><tr><td>Focuses on quantity of ideas</td><td>Focuses on creativity and collaboration</td></tr><tr><td>Requires minimal preparation</td><td>Requires planning and facilitation</td></tr><tr><td>Suitable for small or large groups</td><td>Best for collaborative team environments</td></tr></table> <b>3. Suitable Scenarios with Examples (3 Marks)</b>                                                                                                                                                                                                                                                                                                                                | Brainstorming | Gamestorming | Free-flow idea generation | Uses structured games and activities | Simple and quick to conduct | More interactive and engaging | Focuses on quantity of ideas | Focuses on creativity and collaboration | Requires minimal preparation | Requires planning and facilitation | Suitable for small or large groups | Best for collaborative team environments | CO1 | 1B | K4 |
| Brainstorming                      | Gamestorming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| Free-flow idea generation          | Uses structured games and activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| Simple and quick to conduct        | More interactive and engaging                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| Focuses on quantity of ideas       | Focuses on creativity and collaboration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| Requires minimal preparation       | Requires planning and facilitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |
| Suitable for small or large groups | Best for collaborative team environments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |              |                           |                                      |                             |                               |                              |                                         |                              |                                    |                                    |                                          |     |    |    |

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|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------|------------|---------------|-------------------------|----------------------------|------------------------|----------------|---------------|-----------------------------------|-------------------|------------------------|-----|----|----|
|                           | <p><b>Brainstorming Example (1.5 Marks):</b></p> <ul style="list-style-type: none"><li>• <b>Scenario:</b> Generating ideas for new features in a mobile banking app.</li><li>• Team members quickly suggest ideas such as biometric login, spending tracker, and bill reminders.</li></ul> <p><b>Gamestorming Example (1.5 Marks):</b></p> <ul style="list-style-type: none"><li>• <b>Scenario:</b> Redesigning an e-commerce website.</li><li>• Designers use activities like <b>Crazy 8s</b>, <b>Role Playing</b>, or <b>Empathy Mapping</b> to explore different user perspectives and interface solutions.</li></ul> <p><b>4. Benefits of Integrating Both Techniques (3 Marks)</b></p> <ul style="list-style-type: none"><li>• Encourages <b>creative and diverse ideas</b>.</li><li>• Improves <b>team collaboration and communication</b>.</li><li>• Helps identify <b>innovative and user-centered solutions</b>.</li><li>• Increases <b>participant engagement and motivation</b>.</li><li>• Produces ideas that are both <b>creative and practical</b> for UI/UX design.</li></ul> <p><b>5. Conclusion (1 Mark)</b></p> <ul style="list-style-type: none"><li>• Brainstorming and gamestorming are complementary ideation techniques.</li><li>• Brainstorming promotes rapid idea generation, while gamestorming enhances creativity through structured activities.</li><li>• Combining both techniques leads to innovative, user-focused, and effective UI/UX design solutions.</li></ul>       |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| 4.                        | <p><b>Distinguish between divergent and convergent thinking in problem-solving. How do these thinking styles align with specific stages of Design Thinking, and how can they be effectively applied to enhance the overall user experience in UI/UX design?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO1                       | 1B                         | K5         |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| 4<br>Ans                  | <p><b>1. Introduction (2 Marks)</b></p> <ul style="list-style-type: none"><li>• <b>Divergent Thinking:</b> Generate <b>many ideas</b>, creative, no judgment.</li><li>• <b>Convergent Thinking:</b> <b>Analyze &amp; select</b> the best practical solution.</li></ul> <p><b>2. Differences (4 Marks)</b></p> <table><tr><td><b>Divergent Thinking</b></td><td><b>Convergent Thinking</b></td></tr><tr><td>Many ideas</td><td>Best solution</td></tr><tr><td>Creativity &amp; innovation</td><td>Analysis &amp; decision-making</td></tr><tr><td>Multiple possibilities</td><td>Narrow options</td></tr><tr><td>No evaluation</td><td>Evaluate feasibility &amp; user needs</td></tr><tr><td>Quantity of ideas</td><td>Quality &amp; practicality</td></tr></table> <p><b>3. Design Thinking Stages (3 Marks)</b></p> <ul style="list-style-type: none"><li>• <b>Divergent: Empathize + Ideate</b> → Explore needs, brainstorm.</li><li>• <b>Convergent: Define + Prototype + Test</b> → Refine, evaluate, improve.</li></ul> <p><b>4. UI/UX Application (3 Marks)</b></p> <ul style="list-style-type: none"><li>• <b>Divergent:</b> Multiple layouts, navigation, interaction ideas.<ul style="list-style-type: none"><li>◦ <i>Example:</i> Several fitness app dashboard designs.</li></ul></li><li>• <b>Convergent:</b> Test prototypes, choose best design.<ul style="list-style-type: none"><li>◦ <i>Example:</i> Select checkout flow with fewer errors &amp; faster completion.</li></ul></li></ul> | <b>Divergent Thinking</b> | <b>Convergent Thinking</b> | Many ideas | Best solution | Creativity & innovation | Analysis & decision-making | Multiple possibilities | Narrow options | No evaluation | Evaluate feasibility & user needs | Quantity of ideas | Quality & practicality | CO1 | 1B | K5 |
| <b>Divergent Thinking</b> | <b>Convergent Thinking</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| Many ideas                | Best solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| Creativity & innovation   | Analysis & decision-making                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| Multiple possibilities    | Narrow options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| No evaluation             | Evaluate feasibility & user needs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |
| Quantity of ideas         | Quality & practicality                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |                            |            |               |                         |                            |                        |                |               |                                   |                   |                        |     |    |    |

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|          | <ul style="list-style-type: none"> <li>• <b>Both Together:</b> Creative + practical + user-friendly UI/UX.</li> </ul> <b>5. Conclusion (1 Mark)</b> <ul style="list-style-type: none"> <li>• Divergent = Creativity</li> <li>• Convergent = Practicality</li> <li>• Balance both → Innovative, effective, user-centered UI/UX.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |    |    |
| 5.       | <b>Explore the role of observational empathy in the design process. How does observing user behavior contribute to the creation of meaningful and user-centric UI/UX designs? Provide examples of how observational empathy has been successfully applied in real-world design projects.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO1 | 1B | K4 |
| 5<br>Ans | <b>1. Introduction (2 Marks)</b> <ul style="list-style-type: none"> <li>• <b>Observational Empathy:</b> Understand users by <b>observing real behavior</b>.</li> <li>• Part of <b>Empathize</b> stage in Design Thinking.</li> <li>• Identifies <b>actual user needs</b>, not assumptions.</li> </ul> <b>2. Role in Design Process (4 Marks)</b> <ul style="list-style-type: none"> <li>• Understand <b>behavior, emotions, goals, challenges</b>.</li> <li>• Find <b>hidden problems &amp; pain points</b>.</li> <li>• Supports <b>evidence-based decisions</b>.</li> <li>• Creates <b>user-centered solutions</b>.</li> </ul> <b>3. Contribution to UI/UX (3 Marks)</b> <ul style="list-style-type: none"> <li>• Improve <b>navigation &amp; task flow</b>.</li> <li>• Simplify interactions, reduce errors.</li> <li>• Enhance <b>accessibility &amp; usability</b>.</li> <li>• Create <b>intuitive interfaces</b>.</li> <li>• Increase <b>satisfaction &amp; efficiency</b>.</li> <li>• <b>Example:</b> Move search bar to top after observing users struggle to find it.</li> </ul> <b>4. Real-World Examples (3 Marks)</b> <ul style="list-style-type: none"> <li>• <b>Airbnb:</b> Simpler booking, better listings, improved communication.</li> <li>• <b>IDEO Shopping Cart:</b> Safer, ergonomic, user-friendly carts.</li> <li>• <b>Google Maps:</b> Clearer routes, larger buttons, real-time traffic.</li> </ul> <b>5. Conclusion (1 Mark)</b> <ul style="list-style-type: none"> <li>• <b>Observational Empathy = Real user insights</b></li> <li>• Improves <b>usability, accessibility, and UX</b>.</li> <li>• Ensures designs meet <b>actual user needs</b>.</li> </ul> | CO1 | 1B | K4 |
| 6.       | <b>Illustrate the process of creating a User Journey Map with a suitable example. Explain each stage involved.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO1 | 1B | K3 |

| 6<br>Ans            | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"><li>Brainstorming: Free idea generation, no criticism.</li><li>Gamestorming: Structured games/activities for creativity &amp; collaboration.</li></ul> <p><b>2. Comparison (4 Marks)</b></p> <table><tr><th>Brainstorming</th><th>Gamestorming</th></tr><tr><td>Free-flow ideas</td><td>Structured activities</td></tr><tr><td>Quick &amp; simple</td><td>Interactive &amp; engaging</td></tr><tr><td>Quantity of ideas</td><td>Creativity &amp; collaboration</td></tr><tr><td>Minimal preparation</td><td>Needs planning</td></tr><tr><td>Small/Large groups</td><td>Best for team collaboration</td></tr></table> <p><b>3. Examples (3 Marks)</b></p> <ul style="list-style-type: none"><li>Brainstorming: Mobile banking app → biometric login, spending tracker, bill reminders.</li><li>Gamestorming: E-commerce redesign → Crazy 8s, Role Playing, Empathy Mapping.</li></ul> <p><b>4. Benefits (3 Marks)</b></p> <ul style="list-style-type: none"><li>More creative &amp; diverse ideas.</li><li>Better team collaboration.</li><li>User-centered, innovative solutions.</li><li>Higher engagement &amp; motivation.</li><li>Practical UI/UX improvements.</li></ul> <p><b>5. Conclusion (1 Mark)</b></p> <ul style="list-style-type: none"><li>Brainstorming = Fast idea generation</li><li>Gamestorming = Structured creativity</li><li>Together → Innovative, practical, user-focused UI/UX.</li></ul> | Brainstorming | Gamestorming | Free-flow ideas | Structured activities | Quick & simple | Interactive & engaging | Quantity of ideas | Creativity & collaboration | Minimal preparation | Needs planning | Small/Large groups | Best for team collaboration | CO1 | 1B | K3 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------|-----------------|-----------------------|----------------|------------------------|-------------------|----------------------------|---------------------|----------------|--------------------|-----------------------------|-----|----|----|
| Brainstorming       | Gamestorming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| Free-flow ideas     | Structured activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| Quick & simple      | Interactive & engaging                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| Quantity of ideas   | Creativity & collaboration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| Minimal preparation | Needs planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| Small/Large groups  | Best for team collaboration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |              |                 |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| 7.                  | <p><b>Explain the components of a User Journey Map. Discuss how journey mapping improves customer experience.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO1           | 1B           | K5              |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |
| 7<br>Ans            | <p><b>1. Introduction (2 Marks)</b></p> <ul style="list-style-type: none"><li>User Journey Map: Visual representation of user steps to achieve a goal.</li><li>Helps identify experience, pain points, and improvements.</li></ul> <p><b>2. Components (6 Marks)</b></p> <ol style="list-style-type: none"><li>User Persona: Target user (needs, goals, behavior).</li><li>User Goal: Objective to achieve.</li><li>Journey Stages: Discover → Search → Purchase → Use → Support.</li><li>Touchpoints: Website, app, payment, support.</li><li>Actions, Thoughts &amp; Emotions: What users do, think, feel.</li><li>Pain Points &amp; Opportunities: Problems + improvement areas.</li></ol> <p><b>3. Benefits (4 Marks)</b></p> <ul style="list-style-type: none"><li>Identify pain points.</li><li>Improve usability &amp; interactions.</li><li>Increase customer satisfaction.</li><li>Create user-centered experiences.</li><li>Better team collaboration.</li><li>Improve loyalty &amp; engagement.</li><li>Example: Simplify checkout + add payment options → More successful orders.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                   | CO1           | 1B           | K5              |                       |                |                        |                   |                            |                     |                |                    |                             |     |    |    |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |    |    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
|          | <b>4. Conclusion (1 Mark)</b> <ul style="list-style-type: none"> <li>Journey Mapping = Better understanding of user experience.</li> <li>Improves usability, satisfaction, and UI/UX design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |    |    |
| 8.       | <b>Construct a User Journey Map for an online food delivery application and identify user pain points and improvement opportunities.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO1 | 1B | K4 |
| 8<br>Ans | <b>1. Concept (2 Marks)</b> <ul style="list-style-type: none"> <li><b>User Journey Mapping (UJM):</b> Visual map of user interactions with a product.</li> <li>Shows actions, thoughts, emotions, pain points.</li> <li>Helps improve overall UX.</li> </ul> <b>2. Stages (7 Marks)</b> <ol style="list-style-type: none"> <li><b>User Persona</b> – Target user.</li> <li><b>User Goal</b> – What user wants.</li> <li><b>User Actions</b> – Steps/tasks.</li> <li><b>Touchpoints</b> – Interaction points.</li> <li><b>Pain Points</b> – Problems/frustrations.</li> <li><b>Improve Experience</b> – Better usability &amp; design.</li> </ol> <b>3. Identifying Pain Points (4 Marks)</b> <ul style="list-style-type: none"> <li>Confusing navigation.</li> <li>Long loading/wait times.</li> <li>Unnecessary steps.</li> <li>Process abandonment.</li> <li>User frustration &amp; unmet needs.</li> </ul> <b>4. Enhancing UX</b> <ul style="list-style-type: none"> <li>Simplify workflows.</li> <li>Reduce navigation complexity.</li> <li>Improve accessibility &amp; usability.</li> <li>Increase user satisfaction.</li> <li>Support user-centered design.</li> </ul> <b>5. Example</b> <ul style="list-style-type: none"> <li><b>Food Delivery App:</b> <ul style="list-style-type: none"> <li><b>Goal:</b> Order food.</li> <li><b>Journey:</b> Open App → Search → Select Food → Cart → Payment → Confirmation → Delivery.</li> <li><b>Improvement:</b> One-click payment / Multiple payment options.</li> </ul> </li> </ul> <b>6. Conclusion</b> <ul style="list-style-type: none"> <li><b>UJM = Understand user journey + identify pain points.</b></li> <li><b>Creates intuitive, efficient, user-friendly products.</b></li> </ul> | CO1 | 1B | K4 |

**PART- C**

| Q.No | Questions                                                                                                                                                                                                                                                           | CO's | Unit | Bloom's Level |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1.   | A university intends to redesign its student service portal. a) Create an appropriate User Persona.<br>b) Develop a User Journey Map.<br>c) Identify user pain points.<br>d) Frame an effective problem statement.<br>e) Propose suitable "How Might We" questions. | CO1  | 1C   | K6            |



| 1 Ans        | <p><b>a) Persona (3 Marks)</b></p> <ul style="list-style-type: none"> <li>Name: Rahul / Student</li> <li>Age: 20</li> <li>Goal: Access academic services easily</li> <li>Pain: Confusing portal navigation</li> </ul> <p><b>b) Journey Map (4 Marks)</b></p> <table> <tr> <th>Stage</th> <th>Action</th> <th>Emotion</th> </tr> <tr> <td>Login</td> <td>Enters portal</td> <td>Neutral</td> </tr> <tr> <td>Search</td> <td>Looks for service</td> <td>Confused</td> </tr> <tr> <td>Apply</td> <td>Submits request</td> <td>Frustrated</td> </tr> <tr> <td>Confirmation</td> <td>Receives update</td> <td>Relieved</td> </tr> </table> <p><b>c) Pain Points (3 Marks)</b></p> <ul style="list-style-type: none"> <li>Poor navigation</li> <li>Slow loading pages</li> <li>Lack of clarity in services</li> <li>Complex forms</li> </ul> <p><b>d) Problem Statement (2 Marks)</b></p> <p>“Students struggle to access and complete university services due to a confusing and inefficient portal experience.”</p> <p><b>e) HMW Questions (2 Marks)</b></p> <ul style="list-style-type: none"> <li>How might we simplify navigation in the student portal?</li> <li>How might we reduce time taken to access services?</li> <li>How might we make form submission easier and faster?</li> </ul>  | Stage      | Action | Emotion | Login | Enters portal | Neutral | Search | Looks for service | Confused | Apply | Submits request | Frustrated | Confirmation | Receives update | Relieved | CO1 | 1C | K6 |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|---------|-------|---------------|---------|--------|-------------------|----------|-------|-----------------|------------|--------------|-----------------|----------|-----|----|----|
| Stage        | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Emotion    |        |         |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |
| Login        | Enters portal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Neutral    |        |         |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |
| Search       | Looks for service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Confused   |        |         |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |
| Apply        | Submits request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Frustrated |        |         |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |
| Confirmation | Receives update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Relieved   |        |         |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |
| 2.           | <p><b>A hospital plans to implement a digital appointment management system.</b></p> <ol style="list-style-type: none"> <li>Develop two different User Personas.</li> <li>Construct corresponding User Journey Maps.</li> <li>Frame the key design problems.</li> </ol> <p><b>Recommend appropriate design improvements.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CO1        | 1C     | K6      |       |               |         |        |                   |          |       |                 |            |              |                 |          |     |    |    |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |    |    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
| 2 Ans | <p><b>i) Personas (4 Marks)</b></p> <p><b>Persona 1: Patient (Elderly)</b></p> <ul style="list-style-type: none"> <li>• Age: 65</li> <li>• Needs: Easy appointment booking</li> <li>• Pain: Digital illiteracy</li> </ul> <p><b>Persona 2: Working Professional</b></p> <ul style="list-style-type: none"> <li>• Age: 35</li> <li>• Needs: Quick scheduling</li> <li>• Pain: Time constraints</li> </ul> <p><b>ii) Journey Maps (4 Marks)</b></p> <p><b>Patient Journey</b></p> <ul style="list-style-type: none"> <li>• Search doctor → Confused → Booking → Assisted → Appointment → Satisfied</li> </ul> <p><b>Professional Journey</b></p> <ul style="list-style-type: none"> <li>• Login → Quick search → Book slot → Payment → Reminder → Satisfied</li> </ul> <p><b>PAIN POINTS:</b></p> <ul style="list-style-type: none"> <li>- Complex booking</li> <li>- No reminders</li> <li>- Poor accessibility</li> </ul> <p><b>IMPROVEMENTS:</b></p> <ul style="list-style-type: none"> <li>- 3-step booking</li> <li>- Voice assistant</li> <li>- SMS reminders</li> <li>- Simple UI</li> </ul> <p><b>iii) Design Problems (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Complex booking flow</li> <li>• Poor accessibility for elderly users</li> <li>• Lack of real-time availability</li> <li>• No reminders or notifications</li> </ul> <p><b>iv) Design Improvements (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Simple 3-step booking process</li> <li>• Voice-assisted booking</li> <li>• Real-time slot availability</li> <li>• SMS/email reminders</li> <li>• Mobile-friendly interface</li> <li>• Emergency booking option</li> </ul> | CO1 | 1C | K6 |
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## Unit-II FOUNDATIONS OF UI DESIGN PART-A

| Q.No  | Questions                                                                                                                                                                                    | CO's | Unit | Bloom's Level |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1.    | <b>Why is visual hierarchy important in user interface design?</b>                                                                                                                           | CO2  | 2A   | K1            |
| 1 Ans | Visual hierarchy arranges UI elements according to their importance using size, color, spacing, and contrast. It helps users quickly identify important information and improves navigation. | CO2  | 2A   | K1            |
| 2.    | <b>How is the concept of "contrast" applied in UI design, and why is it essential?</b>                                                                                                       | CO2  | 2A   | K1            |
| 2 Ans | Contrast differentiates elements using color, size, shape, or typography. It improves readability, accessibility, and draws                                                                  | CO2  | 2A   | K1            |

|        |                                                                                                                                                                                             |            |           |           |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|        | attention to important interface components.                                                                                                                                                |            |           |           |
| 3.     | <b>3. Can you give an example of a common UI element used for navigation purposes?</b>                                                                                                      | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 3 Ans  | A <b>navigation bar (Navbar)</b> is a common UI element used to help users move between different pages or sections of an application or website.                                           | CO2        | 2A        | K2        |
| 4.     | <b>What makes the card-based UI pattern valuable in interface design?</b>                                                                                                                   | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 4 Ans  | Card-based UI organizes related information into separate, easy-to-read sections. It improves readability, responsiveness, and content organization across different devices.               | CO2        | 2A        | K1        |
| 5.     | <b>What does "feedback" mean in interaction design, and how does it improve the user experience?</b>                                                                                        | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 5 Ans  | Feedback is the system's response to a user's action, such as notifications or animations. It confirms that an action has been received, reducing confusion and increasing user confidence. | CO2        | 2A        | K2        |
| 6.     | <b>In what ways does the principle of progressive disclosure enhance user interaction?</b>                                                                                                  | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 6 Ans  | Progressive disclosure presents only essential information first and reveals additional details when needed. This reduces complexity and helps users focus on their current task.           | CO2        | 2A        | K2        |
| 7.     | <b>Why is consistency important in reinforcing a brand's identity through design?</b>                                                                                                       | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 7 Ans  | Consistency maintains uniform colors, fonts, icons, and layouts across interfaces. It strengthens brand recognition and provides a familiar user experience.                                | CO2        | 2A        | K1        |
| 8.     | <b>8. How does a brand's personality affect how users perceive and interact with a product?</b>                                                                                             | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 8 Ans  | A brand's personality influences users' emotions, trust, and expectations. A clear personality creates stronger engagement and improves the overall user experience.                        | CO2        | 2A        | K2        |
| 9.     | <b>What is one key advantage of combining a design system with a style guide?</b>                                                                                                           | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 9 Ans  | Combining a design system with a style guide ensures consistent UI components and visual standards, making design and development faster and more efficient.                                | CO2        | 2A        | K1        |
| 10.    | <b>How does the alignment principle help create a visually balanced and effective UI?</b>                                                                                                   | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 10 Ans | Alignment organizes elements in a structured manner, creating a clean and professional layout. It improves readability and guides users through the interface.                              | CO2        | 2A        | K2        |
| 11.    | <b>Give an example of a UI component that promotes user engagement through interactivity.</b>                                                                                               | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 11 Ans | A <b>button</b> is an interactive UI component that allows users to perform actions such as submitting forms, navigating pages, or confirming selections.                                   | CO2        | 2A        | K1        |
| 12.    | <b>How does the principle of affordance influence how users interact with interface elements?</b>                                                                                           | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 12 Ans | Affordance helps users understand how an element should be used through its appearance. For example, raised buttons suggest they                                                            | CO2        | 2A        | K2        |

|           |                                                                                                                                                                                    |            |           |           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|           | can be clicked.                                                                                                                                                                    |            |           |           |
| 13.       | <b>Why is it important to include accessibility standards in a UI design style guide?</b>                                                                                          | <b>CO2</b> | <b>2A</b> | <b>K2</b> |
| 13<br>Ans | Accessibility standards ensure that digital products can be used by people with disabilities. They improve usability, inclusiveness, and compliance with accessibility guidelines. | CO2        | 2A        | K2        |
| 14.       | <b>14. How does implementing a design system support the scalability of a product's interface?</b>                                                                                 | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 14<br>Ans | A design system provides reusable components and consistent design rules. It enables faster updates, easier maintenance, and scalable product development.                         | CO2        | 2A        | K1        |
| 15.       | <b>What role does a brand style guide play in ensuring a unified and consistent visual identity across digital products?</b>                                                       | <b>CO2</b> | <b>2A</b> | <b>K1</b> |
| 15<br>Ans | A brand style guide defines standards for colors, typography, logos, icons, and imagery. It ensures a consistent visual identity across all digital platforms and products.        | CO2        | 2A        | K1        |

### PART-B

| Q.No   | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | COs        | Unit      | Bloom<br>s Level |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|------------------|
| Part B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |           |                  |
| 1.     | <b>Discuss how visual design principles such as alignment, contrast, and hierarchy contribute to enhancing clarity and usability in user interfaces.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO2</b> | <b>2B</b> | <b>K3</b>        |
| 1.Ans  | <b>Introduction (4 Marks)</b> <ul style="list-style-type: none"> <li>Visual Design: Makes UI clear, attractive, usable.</li> <li>Principles: Alignment, Contrast, Visual Hierarchy.</li> <li>Improves clarity, navigation &amp; UX.</li> </ul> <b>1. Alignment (3 Marks)</b> <ul style="list-style-type: none"> <li>Arrange elements in a consistent line.</li> <li>Benefits: Organization, readability, consistency, easy navigation.</li> <li>Example: Username → Password → Login (vertically aligned).</li> </ul> <b>2. Contrast (3 Marks)</b> <ul style="list-style-type: none"> <li>Difference in color, size, brightness, font.</li> <li>Benefits: Readability, attention, accessibility, fewer errors.</li> <li>Example: Orange Buy Now button on white background.</li> </ul> <b>3. Visual Hierarchy (3 Marks)</b> <ul style="list-style-type: none"> <li>Arrange elements by importance.</li> <li>Uses size, color, spacing, position.</li> <li>Benefits: Guides attention, faster decisions, better navigation.</li> <li>Example: Image → Name → Price → Buy Button.</li> </ul> <b>Conclusion</b> <ul style="list-style-type: none"> <li>Alignment = Organization</li> <li>Contrast = Visibility</li> <li>Hierarchy = Priority</li> <li>Together = Better UI &amp; UX</li> </ul> | CO2        | 2B        | K3               |
| 2.     | <b>Explain the role of UI elements and design patterns in creating consistent and user-friendly digital experiences.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO2</b> | <b>2B</b> | <b>K4</b>        |
| 2.Ans  | Introduction <ul style="list-style-type: none"> <li>UI Design: Creates attractive, easy-to-use interfaces.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO2        | 2B        | K4               |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |    |    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
|       | <ul style="list-style-type: none"> <li>• Uses UI Elements + Design Patterns for better UX.</li> </ul> <p>1. UI Elements (3 Marks)</p> <ul style="list-style-type: none"> <li>• Buttons – Actions</li> <li>• Text Fields – Input</li> <li>• Checkbox/Radio – Selection</li> <li>• Dropdown – Multiple options</li> <li>• Icons – Visual actions</li> <li>• Navigation Bar – Move between pages</li> <li>• Slider – Adjust values</li> <li>• Progress Bar – Task progress</li> <li>• Dialog/Popup – Messages</li> </ul> <p>2. Design Patterns (3 Marks)</p> <ul style="list-style-type: none"> <li>• Navigation Menu</li> <li>• Search</li> <li>• Login/Register</li> <li>• Card Layout</li> <li>• Dashboard</li> <li>• Infinite Scroll</li> <li>• Wizard</li> <li>• Pagination</li> </ul> <p>3. Role of UI Elements (2 Marks)</p> <ul style="list-style-type: none"> <li>• Better interaction</li> <li>• Improve usability</li> <li>• Give feedback</li> <li>• Increase accessibility</li> <li>• Easy navigation</li> </ul> <p>4. Role of Design Patterns (2 Marks)</p> <ul style="list-style-type: none"> <li>• Consistency</li> <li>• Faster learning</li> <li>• Better efficiency</li> <li>• Easy development</li> <li>• Higher user satisfaction</li> </ul> <p>5. Examples (3 Marks)</p> <ul style="list-style-type: none"> <li>• Google – Search box.</li> <li>• Instagram – Bottom navigation.</li> <li>• Amazon – Cart, search, product cards.</li> <li>• Banking apps – Wizard pattern.</li> <li>• Microsoft Office – Consistent menus &amp; icons.</li> </ul> |     |    |    |
| 3.    | <b>Analyze how interaction principles like feedback, affordance, and consistency influence user behavior and engagement within interfaces.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO2 | 2B | K4 |
| 3.Ans | <p>Introduction (2 Marks)</p> <ul style="list-style-type: none"> <li>• Interaction Design: Creates effective user-system interaction.</li> <li>• Key principles: Feedback, Affordance, Consistency.</li> <li>• Improves usability, satisfaction, engagement.</li> </ul> <p>1. Feedback (3 Marks)</p> <ul style="list-style-type: none"> <li>• Meaning: System response to user actions.</li> <li>• Examples: Button change, loading spinner, success/error message.</li> <li>• Benefits: <ul style="list-style-type: none"> <li>○ Confirms actions.</li> <li>○ Reduces confusion.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO2 | 2B | K4 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |           |           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|       | <ul style="list-style-type: none"> <li>○ Builds trust.</li> <li>○ Improves engagement.</li> </ul> <p>2. Affordance (3 Marks)</p> <ul style="list-style-type: none"> <li>• Meaning: Design clues showing how to use an element.</li> <li>• Examples: Button = click, search icon = search, slider = drag.</li> <li>• Benefits: <ul style="list-style-type: none"> <li>○ Easy to understand.</li> <li>○ Reduces learning time.</li> <li>○ Prevents errors.</li> <li>○ Supports intuitive use.</li> </ul> </li> </ul> <p>3. Consistency (3 Marks)</p> <ul style="list-style-type: none"> <li>• Meaning: Same layouts, icons, colors, and actions throughout UI.</li> <li>• Examples: Same Save button position, common icons.</li> <li>• Benefits: <ul style="list-style-type: none"> <li>○ Faster learning.</li> <li>○ Less confusion.</li> <li>○ Predictable behavior.</li> <li>○ Builds trust.</li> </ul> </li> </ul> <p>Combined Impact</p> <ul style="list-style-type: none"> <li>• Faster task completion.</li> <li>• Easy learning.</li> <li>• Fewer errors.</li> <li>• Higher confidence &amp; satisfaction.</li> </ul> <p>Examples (2 Marks)</p> <ul style="list-style-type: none"> <li>• Google: Search suggestions (feedback), search bar (affordance), same layout (consistency).</li> <li>• Instagram: Like animation, icons, consistent navigation.</li> <li>• Amazon: Cart notification, cart icon, consistent pages.</li> </ul> |            |           |           |
| 4.    | <b>Describe how visual UI elements can be used to reflect and strengthen a brand's identity across various digital platforms.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>CO2</b> | <b>2B</b> | <b>K3</b> |
| 4.Ans | <p>1. Consistent Brand Experience (3 Marks)</p> <ul style="list-style-type: none"> <li>• Consistency = Strong brand identity.</li> <li>• Use same colors, fonts, logo, layouts.</li> <li>• Unified buttons, forms, navigation.</li> <li>• Example: Apple – Minimal, premium design.</li> </ul> <p>2. Building Trust (2 Marks)</p> <ul style="list-style-type: none"> <li>• Intuitive UX = User trust.</li> <li>• Focus on user needs + easy navigation.</li> <li>• Reduces frustration.</li> <li>• Example: Amazon – Simple shopping flow.</li> </ul> <p>3. Emotional Connection (2 Marks)</p> <ul style="list-style-type: none"> <li>• Design creates feelings and engagement.</li> <li>• Uses color psychology + microinteractions.</li> <li>• Example: Spotify – Colors, animations, personalization.</li> </ul> <p>4. Brand Loyalty (2 Marks)</p> <ul style="list-style-type: none"> <li>• Positive UX encourages repeat users.</li> <li>• Key factors: <ul style="list-style-type: none"> <li>○ Responsive design.</li> <li>○ Easy checkout.</li> <li>○ Smooth experience.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                       | CO2        | 2B        | K3        |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |    |    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
|       | <ul style="list-style-type: none"> <li>• Example: Starbucks app – Ordering, payment, rewards.</li> </ul> <p>5. Memorable Brand Moments (2 Marks)</p> <ul style="list-style-type: none"> <li>• Delight users with animations, rewards, engaging flows.</li> <li>• Creates lasting impressions.</li> <li>• Example: Duolingo – Fun design and feedback.</li> </ul> <p>6. Brand Differentiation (2 Marks)</p> <ul style="list-style-type: none"> <li>• Unique UX/UI helps stand out.</li> <li>• Uses special features + unique design.</li> <li>• Example: Airbnb – Community-focused experience.</li> </ul> <p>Conclusion</p> <ul style="list-style-type: none"> <li>• Good UX/UI = Consistency + Trust + Emotion + Loyalty + Differentiation.</li> <li>• Strong design builds a memorable and recognizable brand.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |    |    |
| 5.    | <b>Discuss the purpose of using a style guide or design system in UI/UX projects and how they support consistency, scalability, and team collaboration.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO2 | 2B | K3 |
| 5.Ans | <p><b>Introduction</b></p> <ul style="list-style-type: none"> <li>• <b>Design System:</b> Collection of reusable components, patterns, and rules for consistent UI.</li> <li>• <b>Style Guide:</b> Document containing visual and design guidelines.</li> <li>• <b>Relation:</b> Design System = Parent, Style Guide = Child.</li> </ul> <p><b>1. Design Systems (5 Marks)</b></p> <ul style="list-style-type: none"> <li>• Maintains <b>consistent design at scale</b>.</li> <li>• Reduces repeated work and improves efficiency.</li> <li>• Creates a <b>shared language</b> between designers and developers.</li> <li>• Contains: <ul style="list-style-type: none"> <li>◦ <b>Component Library</b> – Buttons, forms, UI elements.</li> <li>◦ <b>Pattern Library</b> – Reusable layouts.</li> <li>◦ <b>Style Guides &amp; Resources</b> – Colors, fonts, icons.</li> </ul> </li> <li>• Requires <b>continuous maintenance</b>.</li> </ul> <p><b>2. Style Guides (6 Marks)</b></p> <ul style="list-style-type: none"> <li>• Provides <b>design rules and implementation guidelines</b>.</li> <li>• Types: <ul style="list-style-type: none"> <li>◦ <b>Content Style Guide</b> – Tone, grammar, formatting.</li> <li>◦ <b>Brand Style Guide</b> – Colors, typography, logos.</li> <li>◦ <b>Front-End Style Guide</b> – UI components, layouts, responsive rules.</li> </ul> </li> <li>• Maintained by <b>design, content, and development teams</b>.</li> </ul> <p><b>Difference (Optional)</b></p> <ul style="list-style-type: none"> <li>• <b>Design System:</b> Complete UI framework.</li> <li>• <b>Style Guide:</b> Specific design instructions.</li> </ul> <p><b>Conclusion (2 Mark)</b></p> <ul style="list-style-type: none"> <li>• Design systems and style guides ensure <b>consistent, efficient, and user-friendly UI/UX design</b>.</li> </ul> | CO2 | 2B | K3 |
| 6.    | <b>Examine the consequences of misalignment between brand identity and UI design, and suggest strategies to maintain coherence across touchpoints.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CO2 | 2B | K4 |
| 6.Ans | <p><b>1. Types of Misaligned Branding (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Visual Misalignment:</b> Inconsistent logos, colors, fonts, and design elements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO2 | 2B | K4 |

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|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|       | <ul style="list-style-type: none"> <li>• <b>Message Misalignment:</b> Different or unclear brand values, voice, and communication.</li> </ul> <p><b>2. Consequences (9 Marks)</b></p> <ol style="list-style-type: none"> <li>1. <b>Inconsistent Marketing Assets</b> <ul style="list-style-type: none"> <li>• Different visuals across platforms.</li> <li>• Reduces brand recognition and trust.</li> </ul> </li> <li>2. <b>Brand Fragmentation</b> <ul style="list-style-type: none"> <li>• Creates a disconnected brand image.</li> <li>• Confuses customers.</li> </ul> </li> <li>3. <b>Confused Brand Messaging</b> <ul style="list-style-type: none"> <li>• Brand values and positioning become unclear.</li> <li>• Weakens customer connection.</li> </ul> </li> <li>4. <b>Declining Audience Engagement</b> <ul style="list-style-type: none"> <li>• Confusing messages reduce customer interaction.</li> </ul> </li> <li>5. <b>Weak Brand Position</b> <ul style="list-style-type: none"> <li>• Reduces competitiveness and market relevance.</li> </ul> </li> <li>6. <b>Need for Brand Realignment</b> <ul style="list-style-type: none"> <li>• Reconnect with mission, values, and unique identity.</li> </ul> </li> <li>7. <b>Choosing Right Partners</b> <ul style="list-style-type: none"> <li>• Work with agencies/teams that understand brand goals.</li> </ul> </li> </ol> <p><b>3. Real-World Examples (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Coca-Cola:</b> Maintains consistent colors, logo, and messaging worldwide.</li> <li>• <b>Apple:</b> Consistent minimalist design strengthens brand identity.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• Misaligned branding causes <b>confusion, loss of trust, and reduced loyalty</b>.</li> <li>• Consistent branding creates a <b>strong, recognizable, and successful brand image</b>.</li> </ul> |            |           |           |
| 7.    | <b>Assess how standard UI patterns affect accessibility and outline best practices for ensuring an inclusive and accessible design approach.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>CO2</b> | <b>2B</b> | <b>K4</b> |
| 7.Ans | <p><b>Introduction</b></p> <ul style="list-style-type: none"> <li>• <b>Accessibility:</b> Designing UI for <b>all users</b>, including people with disabilities.</li> <li>• Focus: <b>Inclusion, usability, equal access</b>.</li> </ul> <p><b>1. User Diversity &amp; Inclusion (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Create <b>inclusive designs</b> for different abilities.</li> <li>• Include users with disabilities in <b>user testing</b>.</li> <li>• Identify challenges and improve usability.</li> <li>• Ensures <b>no user is excluded</b>.</li> </ul> <p><b>2. Assistive Technologies (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Design should support tools like: <ul style="list-style-type: none"> <li>◦ <b>Screen readers</b></li> <li>◦ Keyboard navigation</li> <li>◦ Other accessibility tools</li> </ul> </li> <li>• Improves experience for users with impairments.</li> </ul> <p><b>3. Evaluating Accessibility (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Use <b>automated tools + user feedback</b>.</li> <li>• Conduct accessibility testing with real users.</li> <li>• Identify and fix usability barriers.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO 2       | 2B        | K4        |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |    |    |
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|       | <ul style="list-style-type: none"> <li>Continuous improvement ensures accessibility.</li> </ul> <b>4. Responsive &amp; Adaptive Strategies (3 Marks)</b> <ul style="list-style-type: none"> <li>UI adapts to <b>different devices and screen sizes</b>.</li> <li>Use: <ul style="list-style-type: none"> <li>Fluid layouts</li> <li>Media queries</li> </ul> </li> <li>Ensures consistent and accessible experience.</li> </ul> <b>Conclusion</b> <ul style="list-style-type: none"> <li>Accessible UI promotes <b>inclusion and equal access</b>.</li> <li>Testing + adaptive design creates <b>user-friendly experiences for everyone</b>.</li> </ul>                                                                                                                                                                                                                                                                                          |     |    |    |
| 8.    | <b>Explore how emerging trends like dark mode and micro-interactions are influencing traditional UI design principles and evolving design system standards.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CO2 | 2B | K4 |
| 8.Ans | 1. Dark Mode (4 Marks) <ul style="list-style-type: none"> <li>Dark background + light text.</li> <li>Benefits: Less eye strain, battery saving, better focus, accessibility.</li> <li>Challenge: Maintain contrast and readability.</li> </ul> 2. Microinteractions (5 Marks) <ul style="list-style-type: none"> <li>Small animations/feedback during actions.</li> <li>Benefits: Feedback, status updates, user guidance, emotional connection.</li> <li>Examples: Like animation, button effects, form validation.</li> </ul> 3. Voice User Interface (VUI) (4 Marks) <ul style="list-style-type: none"> <li>Interaction using voice commands.</li> <li>Benefits: Hands-free use, accessibility, natural interaction.</li> <li>Needs: Privacy, error handling, language support.</li> </ul> <b>Conclusion</b> <ul style="list-style-type: none"> <li>Modern UI trends improve usability, accessibility, and <i>user engagement</i>.</li> </ul> | CO2 | 2B | K4 |

### PART-C

| Q.No  | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | COs | Unit | Bloom's Level |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|---------------|
|       | Part C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      |               |
| 1     | <b>Explain how combining UI/UX design, Design Thinking, empathy, and ideation techniques helps address complex user challenges. Support your explanation with a real-world example.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO2 | 2C   | K5            |
| 1.ans | <b>Visual Design Principles (7 Marks)</b> <ul style="list-style-type: none"> <li>Principles work together for <b>beauty + usability</b>.</li> <li>1. <b>Balance</b> – Creates stability and organization.</li> <li>2. <b>Contrast</b> – Highlights important elements.</li> <li>3. <b>Alignment</b> – Creates structure and readability.</li> <li>4. <b>Hierarchy</b> – Guides user attention.</li> <li>5. <b>Repetition</b> – Maintains consistency.</li> <li>6. <b>Proximity</b> – Groups related information.</li> <li>7. <b>Whitespace</b> – Improves clarity and reduces clutter.</li> </ul> <b>How They Complement Each Other (3 Marks)</b> <ul style="list-style-type: none"> <li><b>Balance + Alignment</b> → Organized layout.</li> </ul> | CO2 | 2C   | K5            |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |    |    |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
|            | <ul style="list-style-type: none"> <li>• <b>Contrast + Hierarchy</b> → Focus on important content.</li> <li>• <b>Repetition + Proximity</b> → Consistency and better understanding.</li> <li>• <b>Whitespace</b> → Improves clarity and focus.</li> </ul> <p><b>Overall Effectiveness (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Creates <b>attractive and harmonious designs</b>.</li> <li>• Improves <b>navigation and understanding</b>.</li> <li>• Provides <b>consistent user experience</b>.</li> <li>• Enhances <b>communication, engagement, and satisfaction</b>.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• Combined principles create <b>unified, visually appealing, and user-friendly designs</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |    |    |
| 2          | <b>Empathy is considered a fundamental element of UX design. How does it influence decisions related to user interface design?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO2 | 2C | K5 |
| 2. Answers | <p><b>1. Design Thinking (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Empathize:</b> Understand learner issues.</li> <li>• <b>Define:</b> Identify problems.</li> <li>• <b>Ideate:</b> Create solutions.</li> <li>• <b>Prototype:</b> Design mockups.</li> <li>• <b>Test:</b> Improve using feedback.</li> </ul> <p><b>2. Learner Needs (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Easy navigation.</li> <li>• Personalized dashboard.</li> <li>• Search, quizzes, progress tracking.</li> <li>• Clean interface.</li> </ul> <p><b>3. Brand Identity (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Unified style guide:</b> <ul style="list-style-type: none"> <li>◦ Colors, fonts, icons, layouts.</li> </ul> </li> <li>• Creates trust and consistency.</li> </ul> <p><b>4. Accessibility (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• WCAG compliance.</li> <li>• High contrast, captions, screen readers.</li> <li>• Responsive and keyboard-friendly.</li> </ul> <p><b>5. Modern UI/UX (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Gamification:</b> Badges, points, progress.</li> <li>• <b>Responsive Design:</b> All devices.</li> <li>• <b>Micro-interactions:</b> Feedback animations.</li> </ul> <p><b>6. Visual Design (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Balance:</b> Stable layout.</li> <li>• <b>Contrast:</b> Highlight actions.</li> <li>• <b>Hierarchy:</b> Guide attention.</li> <li>• <b>Alignment/Repetition:</b> Consistency.</li> <li>• <b>Whitespace:</b> Clarity.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• Creates an <b>accessible, engaging, and user-friendly learning platform</b>.</li> </ul> | CO2 | 2C | K5 |

### UNIT -III

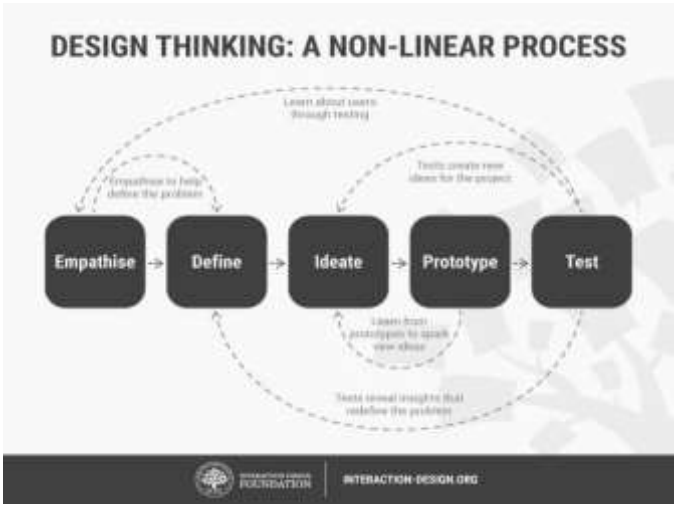
### FOUNDATIONS OF UX DESIGN PART A

| Q.No   | Questions                                                                                                                                                                                   | CO's | Unit | Bloom's Level |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1.     | <b>Define User Experience (UX).</b>                                                                                                                                                         | CO3  | 3A   | K1            |
| 1 Ans  | User Experience (UX) refers to the overall experience a person has when interacting with a product, service, or system, encompassing aspects of usability, accessibility, and satisfaction. | CO3  | 3A   | K1            |
| 2.     | <b>What are the consequences of neglecting User Experience in product development?</b>                                                                                                      | CO3  | 3A   | K1            |
| 2 Ans  | Neglecting User Experience can result in user frustration, decreased user engagement, and potentially, product failure in the market.                                                       | CO3  | 3A   | K1            |
| 3.     | <b>Explain the concept of empathy in UX design.</b>                                                                                                                                         | CO3  | 3A   | K2            |
| 3 Ans  | Empathy involves understanding and sharing the feelings and perspectives of users, allowing designers to create products that truly meet their needs.                                       | CO3  | 3A   | K2            |
| 4.     | <b>How does User Experience differ from User Interface?</b>                                                                                                                                 | CO3  | 3A   | K1            |
| 4 Ans  | User Experience (UX) is a broader concept that includes the overall experience, while User Interface (UI) specifically deals with the visual elements and interactions of a product.        | CO3  | 3A   | K1            |
| 5.     | <b>Why is an iterative approach crucial in UX design?</b>                                                                                                                                   | CO3  | 3A   | K2            |
| 5 Ans  | Iteration allows designers to refine and improve the design based on user feedback, ensuring that the final product meets user needs effectively.                                           | CO3  | 3A   | K2            |
| 6.     | <b>What is the purpose of user research in UX design?</b>                                                                                                                                   | CO3  | 3A   | K1            |
| 6 Ans  | User research helps designers gain insights into user behavior, preferences, and pain points, informing the design process.                                                                 | CO3  | 3A   | K1            |
| 7.     | <b>Name a commonly used tool for creating UX wireframes.</b>                                                                                                                                | CO3  | 3A   | K1            |
| 7 Ans  | Balsamiq is a popular tool for creating low-fidelity wireframes in UX design.                                                                                                               | CO3  | 3A   | K1            |
| 8.     | <b>How can heatmaps be useful in UX research?</b>                                                                                                                                           | CO3  | 3A   | K1            |
| 8 Ans  | Heatmaps visually represent user interactions with a webpage, helping designers understand where users focus their attention.                                                               | CO3  | 3A   | K1            |
| 9.     | <b>What is the significance of defining user personas in UX design?</b>                                                                                                                     | CO3  | 3A   | K1            |
| 9 Ans  | User personas help create a detailed understanding of the target audience, informing design decisions based on user needs and goals.                                                        | CO3  | 3A   | K1            |
| 10.    | <b>How can user stories aid in the UX design process?</b>                                                                                                                                   | CO3  | 3A   | K1            |
| 10 Ans | User stories help capture specific features or functionalities from the user's perspective, guiding the development and                                                                     | CO3  | 3A   | K1            |

|        |                                                                                                                                                                 |            |           |           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|        | design process.                                                                                                                                                 |            |           |           |
| 11.    | <b>How can UX design impact a company's bottom line?</b>                                                                                                        | <b>CO3</b> | <b>3A</b> | <b>K1</b> |
| 11 Ans | Positive UX design can lead to increased customer satisfaction and loyalty, ultimately contributing to higher revenue and profitability.                        | CO3        | 3A        | K1        |
| 12.    | <b>Explain the concept of usability in User Experience.</b>                                                                                                     | <b>CO3</b> | <b>3A</b> | <b>K2</b> |
| 12 Ans | Usability refers to the ease with which users can interact with a product, emphasizing efficiency, learnability, and user satisfaction.                         | CO3        | 3A        | K2        |
| 13.    | <b>How does User Experience extend beyond digital products?</b>                                                                                                 | <b>CO3</b> | <b>3A</b> | <b>K1</b> |
| 13 Ans | User Experience extends beyond digital products to include experiences with services, environments, and any interaction users have with a brand or organization | CO3        | 3A        | K1        |
| 14.    | <b>In what ways can a positive User Experience contribute to customer retention?</b>                                                                            | <b>CO3</b> | <b>3A</b> | <b>K1</b> |
| 14 Ans | Positive User Experiences create a sense of satisfaction and trust, encouraging customers to continue using a product or service.                               | CO3        | 3A        | K1        |
| 15.    | <b>How might a seamless User Experience positively influence a company's reputation?</b>                                                                        | <b>CO3</b> | <b>3A</b> | <b>K1</b> |
| 15 Ans | A seamless User Experience can lead to positive reviews, recommendations, and an enhanced reputation, attracting more users to engage with the brand.           | CO3        | 3A        | K1        |

#### PART- B

| Q.No  | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO's       | Unit      | Bloom's Level |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|---------------|
| 1.    | <b>Explain the concept of User Experience (UX) and its significance in today's digital landscape.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>CO3</b> | <b>3B</b> | <b>K3</b>     |
| 1 Ans | <b>Definition (2 Marks)</b> <ul style="list-style-type: none"> <li>UX: Overall user experience while using a product, website, or app.</li> <li>Focuses on usefulness, usability, accessibility, and satisfaction.</li> </ul> <b>Components of UX (5 Marks)</b> <ol style="list-style-type: none"> <li>Usability – Easy to learn and use.</li> <li>Accessibility – Supports all users, including disabilities.</li> <li>Visual Design – Attractive and engaging interface.</li> <li>Interaction Design – Simple buttons, menus, navigation.</li> <li>Information Architecture – Logical content organization.</li> <li>Performance – Fast and responsive system.</li> </ol> <b>Significance of UX (6 Marks)</b> <ol style="list-style-type: none"> <li>Customer Satisfaction – Better user experience.</li> <li>User Retention – Encourages repeat use.</li> <li>Brand Reputation – Builds trust.</li> <li>Revenue Growth – Improves sales and conversions.</li> </ol> | <b>CO3</b> | <b>3B</b> | <b>K3</b>     |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |           |           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|       | 5. Cost Reduction – Prevents redesign issues.<br>6. Competitive Advantage – Helps stand out.<br><b>Conclusion</b> <ul style="list-style-type: none"> <li>Good UX creates simple, efficient, and satisfying user experiences.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |           |           |
| 2.    | <b>Explain the UX design process and its methodology.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CO3</b> | <b>3B</b> | <b>K3</b> |
| 2 Ans | <b>Definition (2 Marks)</b> <ul style="list-style-type: none"> <li>UX Design Process: A structured, iterative method to create user-centered solutions.</li> <li>Includes research, ideation, prototyping, and testing.</li> </ul> <b>Stages (5Marks)</b> <ol style="list-style-type: none"> <li>Empathize           <ul style="list-style-type: none"> <li>Understand users through interviews, observation, research.</li> </ul> </li> <li>Define           <ul style="list-style-type: none"> <li>Identify problems using user insights, personas, journey maps.</li> </ul> </li> <li>Ideate           <ul style="list-style-type: none"> <li>Generate multiple ideas through brainstorming and creativity.</li> </ul> </li> <li>Prototype           <ul style="list-style-type: none"> <li>Create sketches/mockups to test ideas.</li> </ul> </li> <li>Test           <ul style="list-style-type: none"> <li>Collect user feedback and improve solutions.</li> </ul> </li> </ol> <b>Key Characteristics (4 Marks)</b> <ul style="list-style-type: none"> <li><b>Human-centered – Focus on user needs.</b></li> <li>Iterative – Continuous improvement.</li> <li>Collaborative – Team-based process.</li> <li>Evidence-based – Uses research and testing</li> </ul> <b>Diagram : (2 Marks)</b>  | <b>CO3</b> | <b>3B</b> | <b>K3</b> |
| 3.    | <b>Discuss why businesses and designers alike should prioritize and invest in creating a positive user experience. Provide examples to support your arguments.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CO3</b> | <b>3B</b> | <b>K4</b> |
| 3 Ans | <b>Introduction (2 Marks)</b> <ul style="list-style-type: none"> <li>UX is a business and design priority.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>CO3</b> | <b>3B</b> | <b>K4</b> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |           |           |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|
|       | <ul style="list-style-type: none"> <li>• Good UX improves user satisfaction and product success.</li> </ul> <p><b>Business Justification (5 Marks)</b></p> <ol style="list-style-type: none"> <li>1. <b>Higher Conversions – Simple flows increase sales/sign-ups.</b></li> <li>2. Customer Retention – Good UX reduces user drop-off.</li> <li>3. Lower Support Costs – Fewer user problems.</li> <li>4. Word-of-Mouth Growth – Happy users recommend products.</li> <li>5. Competitive Advantage – Better experience attracts users.</li> </ol> <p><b>Designer Justification (3 Marks)</b></p> <ol style="list-style-type: none"> <li>1. User Needs – Solves real problems.</li> <li>2. Ethical &amp; Inclusive Design – Supports diverse users.</li> <li>3. Data-Based Decisions – Uses research and testing.</li> </ol> <p><b>Real-World Examples (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Airbnb: Better photos + booking UX increased trust.</li> <li>• Amazon: 1-click checkout reduced purchase friction.</li> <li>• Slack: Simple onboarding improved adoption.</li> <li>• Google: Simple search interface built user trust.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• Good UX creates satisfied users, business growth, and better products.</li> </ul> |            |           |           |
| 4.    | <b>How do UX design process and its methodology contribute to creating user centric products and services?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>CO3</b> | <b>3B</b> | <b>K4</b> |
| 4 Ans | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• UX process keeps users at the center of design decisions.</li> <li>• Uses research, testing, and iteration to create user-focused products.</li> </ul> <p><b>Key Contributions (9 Marks)</b></p> <ol style="list-style-type: none"> <li>1. Evidence-Based Decisions</li> <li>• Uses real user research, not assumptions.</li> <li>2. Right Problem Identification</li> <li>• Define stage finds the actual user problem.</li> <li>3. Multiple Solutions</li> <li>• Ideation encourages creative options.</li> <li>4. Early Validation</li> <li>• Prototypes test ideas before development.</li> <li>5. Continuous Improvement</li> <li>• Testing and iteration refine products.</li> <li>6. Accessibility &amp; Inclusion</li> <li>• Identifies needs of diverse users.</li> <li>7. Business + User Goals</li> <li>• Improves satisfaction, retention, and growth.</li> </ol> <p><b>Conclusion (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• UX methodology creates adaptive, evidence-based, user-centered solutions.</li> <li>• Continuous feedback ensures products meet real user needs.</li> </ul>                                                                                                                     | <b>CO3</b> | <b>3B</b> | <b>K4</b> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |           |           |
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| 5.    | <b>Explore the role of research in User Experience Design.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CO3</b> | <b>3B</b> | <b>K3</b> |
| 5 Ans | <p><b>Introduction (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>UX Research:</b> Systematic study of users, needs, behavior, and motivations.</li> <li>• Helps make <b>evidence-based design decisions</b>.</li> <li>• Prevents <b>guesswork</b> and improves user-focused products.</li> </ul> <p><b>Research is Central to UX Design (7 Marks)</b></p> <ol style="list-style-type: none"> <li>1. <b>Finds Real Needs</b> – Replaces assumptions with user data.</li> <li>2. <b>Validates Ideas</b> – Tests design concepts before investment.</li> <li>3. <b>Improves Information Architecture</b> – Creates user-friendly navigation.</li> <li>4. <b>Builds Personas &amp; Journey Maps</b> – Based on real users.</li> <li>5. <b>Reduces Rework</b> – Finds problems early.</li> <li>6. <b>Prioritizes Features</b> – Focuses on user value.</li> <li>7. <b>Measures Success</b> – Uses task success, errors, time metrics.</li> </ol> <p><b>Research Cycle:</b></p> <ul style="list-style-type: none"> <li>• Research → Design → Prototype → Test → Improve.</li> </ul> <p><b>Types of Research (3 Marks)</b></p> <p><b>1. Generative Research</b></p> <ul style="list-style-type: none"> <li>• Finds problems and user needs.</li> <li>• Methods: Interviews, surveys, observation.</li> <li>• Answers: <b>“What should we build?”</b></li> </ul> <p><b>2. Evaluative Research</b></p> <ul style="list-style-type: none"> <li>• Tests designs and products.</li> <li>• Methods: Usability testing, A/B testing.</li> <li>• Answers: <b>“Is it working?”</b></li> </ul> | <b>CO3</b> | <b>3B</b> | <b>K3</b> |
| 6.    | <b>Discuss various research methods and tools used in UX research.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CO3</b> | <b>3B</b> | <b>K5</b> |
|       | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• UX research has two types: <ul style="list-style-type: none"> <li>◦ Qualitative: Explains <i>why</i> users behave.</li> <li>◦ Quantitative: Measures <i>what/how much</i>.</li> </ul> </li> <li>• Combining both gives complete user insights.</li> </ul> <p><b>Qualitative Methods (4 Marks)</b></p> <ol style="list-style-type: none"> <li>1. User Interviews – Understand needs, feelings, and problems.</li> <li>2. Contextual Inquiry – Observe users in real environments.</li> <li>3. Usability Testing – Test designs by observing user tasks.</li> <li>4. Focus Groups &amp; Diary Studies – Collect opinions and long-term behavior.</li> </ol> <p><b>Quantitative Methods (4 Marks)</b></p> <ol style="list-style-type: none"> <li>1. Surveys – Collect large-scale user feedback.</li> <li>2. A/B Testing – Compare design versions.</li> <li>3. Analytics &amp; Heatmaps – Track user actions.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CO3</b> | <b>3B</b> | <b>K5</b> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |    |    |
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|       | <p>4. Card Sorting &amp; Tree Testing – Improve navigation structure.</p> <p><b>Common Tools (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Surveys: Google Forms, Typeform.</li> <li>• Usability Testing: UserTesting, Maze.</li> <li>• Analytics: Hotjar, Google Analytics.</li> <li>• Prototyping: Figma, Adobe XD.</li> <li>• Remote Interviews: Zoom, Microsoft Teams.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• Qualitative + Quantitative research creates data-driven, user-centered UX designs.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |    |    |
| 7.    | <p><b>How does user research contribute to the effectiveness of the design process, ensuring that user needs are understood and addressed?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO3 | 3B | K4 |
| 7 Ans | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• User Research: Connects design decisions with real user needs.</li> <li>• Reduces assumptions and creates effective user-centered products.</li> </ul> <p><b>Key Contributions (9 Marks)</b></p> <ol style="list-style-type: none"> <li>1. Validates Assumptions – Confirms real user needs before development.</li> <li>2. Finds Problems Early – Reduces costly changes after launch.</li> <li>3. Improves Personas &amp; Journey Maps – Based on real user data.</li> <li>4. Prioritizes Features – Focuses on important user needs.</li> <li>5. Provides Metrics – Measures task success, errors, satisfaction.</li> <li>6. Aligns Teams – Supports decisions with research evidence.</li> </ol> <p><b>User Needs Understanding (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Continuous research keeps designs connected to users.</li> <li>• Creates a feedback loop: Research → Design → Test → Improve.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>• User research ensures accurate, effective, and user-focused UX design.</li> </ul> | CO3 | 3B | K4 |
| 8.    | <p><b>Provide Examples of How Research Insights Have Influenced Successful UX Design Outcomes</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO3 | 3B | K5 |
| 8 Ans | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Research reveals real user problems.</li> <li>• Helps create better design decisions and improvements.</li> </ul> <p><b>1. Airbnb – Photography (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Research found poor listing photos reduced trust.</li> <li>• Professional photos increased bookings.</li> <li>• Insight → Better trust + growth.</li> </ul> <p><b>2. Spotify – Listening Data (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Studied user behavior and preferences.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO3 | 3B | K5 |



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|  | <ul style="list-style-type: none"> <li>Created Discover Weekly recommendations.</li> <li>Increased engagement and retention.</li> </ul> <p><b>3. Microsoft Office – Ribbon Design (2 Marks)</b></p> <ul style="list-style-type: none"> <li>Research showed users struggled to find tools.</li> <li>Introduced Ribbon interface.</li> <li>Improved task efficiency.</li> </ul> <p><b>4. Google – Simple Search (2 Marks)</b></p> <ul style="list-style-type: none"> <li>Research found clutter reduced usability.</li> <li>Minimal homepage improved speed and trust.</li> </ul> <p><b>5. Slack – Onboarding (2 Marks)</b></p> <ul style="list-style-type: none"> <li>Research identified user confusion.</li> <li>Slackbot guided new users.</li> <li>Improved adoption and activation.</li> </ul> <p><b>Conclusion</b></p> <ul style="list-style-type: none"> <li>User research converts insights into effective designs and improves UX success.</li> </ul> |  |  |  |
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### PART- C

| Q.No  | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO's       | Unit      | Bloom's Level |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|---------------|
| 1.    | <b>In what ways does aligning UX design with business objectives influence the success and strategic value of digital products?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CO3</b> | <b>3C</b> | <b>K6</b>     |
| 1 Ans | <p><b>Introduction (2 Marks)</b></p> <ul style="list-style-type: none"> <li>UX and business goals must work together for product success.</li> <li>Good UX improves user satisfaction, revenue, and growth.</li> </ul> <p><b>Key Benefits of Alignment (8 Marks)</b></p> <ol style="list-style-type: none"> <li>Higher Conversion Rate <ul style="list-style-type: none"> <li>Simple flows and clear CTAs improve sales and sign-ups.</li> </ul> </li> <li>Customer Retention <ul style="list-style-type: none"> <li>Good UX increases engagement and customer lifetime value.</li> </ul> </li> <li>Lower Acquisition Cost <ul style="list-style-type: none"> <li>Happy users promote products through referrals.</li> </ul> </li> <li>Better Decision Making <ul style="list-style-type: none"> <li>Research-based UX reduces wrong investments.</li> </ul> </li> <li>Competitive Advantage <ul style="list-style-type: none"> <li>Superior experience strengthens brand identity.</li> </ul> </li> <li>Risk Reduction <ul style="list-style-type: none"> <li>Prevents building unnecessary features and saves resources.</li> </ul> </li> </ol> <p><b>Example: Duolingo (2 Marks)</b></p> <ul style="list-style-type: none"> <li>Uses streaks, rewards, and reminders to increase daily usage.</li> <li>UX supports retention and paid subscriptions.</li> </ul> | <b>CO3</b> | <b>3C</b> | <b>K6</b>     |

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|       | <b>Example: Netflix (2 Marks)</b> <ul style="list-style-type: none"> <li>Personalized recommendations improve content discovery.</li> <li>Better UX reduces churn and increases subscriptions.</li> </ul> <b>Conclusion</b> <ul style="list-style-type: none"> <li>Aligning UX with business goals creates products that are user-friendly, profitable, and successful.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |    |    |
| 2.    | <b>Evaluate the effectiveness of various tools and methods used for research in User Experience Design.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO3 | 3C | K6 |
| 2 Ans | <b>Introduction (2 Marks)</b> <ul style="list-style-type: none"> <li>UX research methods are evaluated based on insight depth, scalability, cost, reliability, and suitability.</li> <li>No single method is best; effectiveness depends on the situation.</li> </ul> <b>Evaluation Criteria</b> <ul style="list-style-type: none"> <li>Depth of Insight – Understanding user behavior and reasons.</li> <li>Scalability – Number of users reached.</li> <li>Cost &amp; Time – Resources required.</li> <li>Reliability – Accuracy of findings.</li> <li>Stage Suitability – Use in discovery or testing.</li> </ul> <b>Qualitative Methods (6 Marks)</b> <ol style="list-style-type: none"> <li>User Interviews <ul style="list-style-type: none"> <li>High depth; reveals user needs and motivations.</li> <li>Limited by small samples and bias.</li> <li>Best for early research.</li> </ul> </li> <li>Contextual Inquiry <ul style="list-style-type: none"> <li>Observes users in real environments.</li> <li>Finds hidden behaviors.</li> <li>Time-consuming and costly.</li> </ul> </li> <li>Usability Testing <ul style="list-style-type: none"> <li>Identifies design problems directly.</li> <li>Effective for prototype evaluation.</li> <li>Limited scalability.</li> </ul> </li> <li>Focus Groups <ul style="list-style-type: none"> <li>Collects opinions quickly.</li> <li>Risk of group bias.</li> </ul> </li> <li>Diary Studies <ul style="list-style-type: none"> <li>Studies long-term user behavior.</li> <li>High dropout risk.</li> </ul> </li> </ol> <b>Quantitative Methods (6 Marks)</b> <ol style="list-style-type: none"> <li>Surveys <ul style="list-style-type: none"> <li>Large-scale data collection.</li> <li>Shows trends but limited "why" insights.</li> </ul> </li> <li>A/B Testing <ul style="list-style-type: none"> <li>Measures impact of design changes.</li> </ul> </li> </ol> | CO3 | 3C | K6 |

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|  | <ul style="list-style-type: none"> <li>Requires enough users and traffic.</li> </ul>                                                                                                     |  |  |  |
|  | 3. Analytics & Heatmaps <ul style="list-style-type: none"> <li>Shows user behavior patterns.</li> <li>Does not explain reasons behind actions.</li> </ul>                                |  |  |  |
|  | 4. Card Sorting & Tree Testing <ul style="list-style-type: none"> <li>Useful for improving navigation and information architecture.</li> <li>Limited to specific UX problems.</li> </ul> |  |  |  |

**UNIT- 4**  
**PART A**

| Q.No  | Questions                                                                                                                                                                           | CO's | Unit | Bloom's Level |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1.    | <b>What is a fundamental principle of effective sketching in UX design, and how can it be applied to improve the design process?</b>                                                | CO4  | 4A   | K1            |
| 1 Ans | Clarity - Sketches should communicate ideas clearly, emphasizing essential elements without unnecessary details.                                                                    | CO4  | 4A   | K1            |
| 2.    | <b>Why sketching is considered a quick and efficient ideation tool in the design process.</b>                                                                                       | CO4  | 4A   | K1            |
| 2 Ans | Sketching allows designers to rapidly explore and communicate ideas, fostering quick iteration and collaboration.                                                                   | CO4  | 4A   | K1            |
| 3.    | <b>Explain the primary focus of sketching red routes in UX design</b>                                                                                                               | CO4  | 4A   | K2            |
| 3 Ans | Identifying and sketching the critical paths or scenarios that users follow, emphasizing the most crucial and frequently used functionalities.                                      | CO4  | 4A   | K2            |
| 4.    | <b>How do sketching red routes contribute to prioritizing design efforts?</b>                                                                                                       | CO4  | 4A   | K1            |
| 4 Ans | It helps prioritize design elements and interactions that are vital to the user experience, ensuring a focus on critical user journeys.                                             | CO4  | 4A   | K1            |
| 5.    | <b>Define responsive design in UX.</b>                                                                                                                                              | CO4  | 4A   | K2            |
| 5 Ans | Responsive design is an approach that ensures a seamless and consistent user experience across various devices and screen sizes.                                                    | CO4  | 4A   | K2            |
| 6.    | <b>Discuss the significance of mobile responsiveness in today's digital landscape and evaluate its impact on user experience and engagement.</b>                                    | CO4  | 4A   | K2            |
| 6 Ans | With the increasing use of mobile devices, responsive design ensures that websites and applications adapt to different screen sizes, improving accessibility and user satisfaction. | CO4  | 4A   | K2            |
| 7.    | <b>Describe the primary purpose of creating wireframes in UX design?</b>                                                                                                            | CO4  | 4A   | K2            |
| 7 Ans | Wireframes serve as low-fidelity representations of a design, outlining the structure and layout of a webpage or application.                                                       | CO4  | 4A   | K2            |
| 8.    | <b>How do wireframes contribute to collaboration in a design team?</b>                                                                                                              | CO4  | 4A   | K1            |
| 8 Ans | Wireframes provide a visual reference that facilitates communication among team members, aligning everyone on the basic structure and functionality of a design.                    | CO4  | 4A   | K1            |
| 9.    | <b>How do wireflows differ from traditional wireframes?</b>                                                                                                                         | CO4  | 4A   | K1            |

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| 9 Ans  | Wireflows combine the structural clarity of wireframes with the sequential flow of user journeys, providing a more comprehensive view of the user experience.           | CO4 | 4A | K1 |
| 10.    | <b>Define the purpose of a prototype in UX design.</b>                                                                                                                  | CO4 | 4A | K2 |
| 10 Ans | A prototype is a tangible representation of a design that allows users to interact with and test the functionality of a product before final development.               | CO4 | 4A | K2 |
| 11.    | <b>Summarize the popular design tool used for creating wireframes and prototypes.</b>                                                                                   | CO4 | 4A | K2 |
| 11 Ans | Sketch is a popular design tool that allows designers to create wireframes and interactive prototypes.                                                                  | CO4 | 4A | K2 |
| 12.    | <b>Discuss How Design Tools Improve Collaboration Among Team Members</b>                                                                                                | CO4 | 4A | K2 |
| 12 Ans | Design tools often support version control and collaboration features, enabling multiple team members to work on the same project simultaneously.                       | CO4 | 4A | K2 |
| 13.    | <b>Describe an example of a common interaction pattern in mobile app design.</b>                                                                                        | CO4 | 4A | K2 |
| 13 Ans | The "Swipe" or "Swipe Left/Right" interaction pattern is often used for navigation or content browsing in mobile applications.                                          | CO4 | 4A | K2 |
| 14.    | <b>Explain How do interaction patterns contribute to a consistent user experience?</b>                                                                                  | CO4 | 4A | K2 |
| 14 Ans | Consistent use of interaction patterns across a product helps users develop familiarity, making the interface more intuitive and user-friendly.                         | CO4 | 4A | K2 |
| 15.    | <b>Explain the purpose of usability testing in UX design</b>                                                                                                            | CO4 | 4A | K2 |
| 15 Ans | Usability testing is conducted to evaluate how easily users can interact with a product and identify areas for improvement in terms of usability and user satisfaction. | CO4 | 4A | K2 |

#### PART- B

| Q.No  | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO's | Unit | Bloom's Level |
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| 1.    | <b>Analyze how removing visual design elements (colors, images, typography) from a wireframe affects the way stakeholders interpret functionality and content hierarchy, and explain why this differs from feedback on a high-fidelity prototype.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO4  | 4B   | K4            |
| 1 Ans | <b>Definition (2 Marks)</b> <ul style="list-style-type: none"> <li>Wireframing is a basic visual layout of a website/app interface.</li> <li>Focuses on structure, functionality, navigation, and content hierarchy without colors or images.</li> </ul> <b>Purpose (2 Marks)</b> <ul style="list-style-type: none"> <li>Concept Visualization: Shows interface structure.</li> <li>Functional Blueprint: Guides designers and developers.</li> <li>User-Centered Design: Improves navigation and UX.</li> <li>Cost-Efficient: Allows quick changes before development.</li> </ul> <b>Key Elements (3 Marks)</b> <ul style="list-style-type: none"> <li>Layout: Arrangement of headers, menus, and content.</li> <li>Content: Placement of text, images, and media.</li> <li>Functionality: Buttons, forms, and interactions.</li> </ul> | CO4  | 4B   | K3            |

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|          | <ul style="list-style-type: none"> <li>• Hierarchy: Shows importance of information.</li> <li>• Navigation: Displays page flow.</li> </ul> <b>Types of Wireframes (3 Marks)</b> <ol style="list-style-type: none"> <li>1. Low-Fidelity <ul style="list-style-type: none"> <li>○ Simple sketches focusing on structure.</li> <li>○ Quick and easy to modify.</li> </ul> </li> <li>2. High-Fidelity <ul style="list-style-type: none"> <li>○ Detailed designs with fonts, colors, and visuals.</li> <li>○ Close to final interface.</li> </ul> </li> </ol> <b>Best Practices (3 Marks)</b> <ul style="list-style-type: none"> <li>• Understand user needs.</li> <li>• Keep design simple.</li> <li>• Improve through feedback and iteration.</li> <li>• Maintain consistency.</li> <li>• Collaborate with designers, developers, and users.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |           |           |
| 2.       | <b>Demonstrate the principles of effective sketching and explain how it contributes to ideation, collaboration, and problem-solving.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>CO4</b> | <b>4B</b> | <b>K4</b> |
| 2<br>Ans | <b>Introduction (2 Marks)</b> <ul style="list-style-type: none"> <li>• Sketching is a basic skill used in art, design, architecture, and engineering.</li> <li>• It helps represent ideas, forms, and concepts visually.</li> </ul> <b>Principles of Sketching (11 Marks)</b> <ol style="list-style-type: none"> <li>1. Observation – Study details, proportions, and relationships.</li> <li>2. Simplification – Break complex objects into basic shapes.</li> <li>3. Line Quality – Use different line weights for depth and texture.</li> <li>4. Proportion &amp; Scale – Maintain accurate size relationships.</li> <li>5. Negative Space – Use empty spaces to improve accuracy.</li> <li>6. Composition – Arrange elements for balance and visual appeal.</li> <li>7. Perspective – Create depth using perspective techniques.</li> <li>8. Contrast &amp; Values – Use light and dark shades for emphasis.</li> <li>9. Texture – Show surface details through lines and shading.</li> <li>10. Consistency – Maintain a unified drawing style.</li> <li>11. Experimentation – Try different techniques and materials.</li> <li>12. Practice – Regular practice improves skill.</li> <li>13. Gesture &amp; Flow – Capture movement and energy.</li> <li>14. Feedback &amp; Learning – Improve through critique and study.</li> <li>15. Creativity &amp; Expression – Add personal style and ideas.</li> </ol> <b>Conclusion</b> <ul style="list-style-type: none"> <li>• Effective sketching requires observation, practice, creativity, and continuous improvement.</li> </ul> | CO4        | 4B        | K4        |
| 3.       | <b>Analyze how user feedback and usability testing drive successive prototype refinements, and explain why this iterative approach better aligns the final product with user needs than a single-pass design approach.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CO4</b> | <b>4B</b> | <b>K4</b> |
| 3<br>Ans | <b>Introduction (3 Marks)</b> <ul style="list-style-type: none"> <li>• Prototype iteration is the continuous improvement of prototypes using user feedback, testing, and requirements.</li> <li>• It ensures the final product meets user needs and expectations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO3        | 4B        | K3        |

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|          | <b>Key Aspects of Prototype Iteration (5 Marks)</b> <ol style="list-style-type: none"> <li>1. Gather User Feedback <ul style="list-style-type: none"> <li>• Collect insights through testing, surveys, and interviews.</li> <li>• Identifies user needs and problems.</li> </ul> </li> <li>2. Identify Design Issues <ul style="list-style-type: none"> <li>• Finds usability problems and design flaws.</li> <li>• Helps create targeted improvements.</li> </ul> </li> <li>3. Prioritize Enhancements <ul style="list-style-type: none"> <li>• Focuses on important issues affecting user experience.</li> <li>• Improves design efficiency.</li> </ul> </li> <li>4. Implement Design Changes <ul style="list-style-type: none"> <li>• Updates features and improves usability.</li> <li>• Aligns prototype with user expectations.</li> </ul> </li> <li>5. Usability Testing <ul style="list-style-type: none"> <li>• Tests updated prototypes.</li> <li>• Measures effectiveness of changes.</li> </ul> </li> <li>6. Continuous Refinement <ul style="list-style-type: none"> <li>• Repeated improvements create a better final product.</li> </ul> </li> </ol> <b>Benefits of Prototype Iteration (5 Marks)</b> <ul style="list-style-type: none"> <li>• User-Centered Design: Matches user needs and increases satisfaction.</li> <li>• Early Problem Detection: Reduces development costs.</li> <li>• Improved Usability: Creates simple and effective interfaces.</li> <li>• Flexibility: Adapts to changing requirements.</li> <li>• Data-Driven Decisions: Uses feedback and testing results for improvements.</li> </ul> |     |    |    |
| 4.       | Given raw data from a usability test, user interviews, and a heuristic evaluation on the same product feature, apply a synthesis process to organize these findings into a set of actionable design insights.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO4 | 4B | K3 |
| 4<br>Ans | <b>Introduction (3 Marks)</b> <ul style="list-style-type: none"> <li>• Synthesizing test findings is the process of analyzing UX research data from usability tests, interviews, surveys, and evaluations.</li> <li>• It converts data into insights for improving design decisions.</li> </ul> <b>Key Steps (5 Marks)</b> <ol style="list-style-type: none"> <li>1. Review Data <ul style="list-style-type: none"> <li>• Analyze user feedback, behaviors, and metrics.</li> </ul> </li> <li>2. Find Patterns <ul style="list-style-type: none"> <li>• Identify common problems and themes.</li> </ul> </li> <li>3. Prioritize Issues <ul style="list-style-type: none"> <li>• Rank problems based on severity and impact.</li> </ul> </li> <li>4. Create Personas &amp; User Journeys <ul style="list-style-type: none"> <li>• Understand user goals, behaviors, and pain points.</li> </ul> </li> <li>5. Affinity Diagrams <ul style="list-style-type: none"> <li>• Group similar findings for better analysis.</li> </ul> </li> <li>6. Empathy Maps <ul style="list-style-type: none"> <li>• Understand user thoughts, feelings, and challenges.</li> </ul> </li> <li>7. Visualize Data <ul style="list-style-type: none"> <li>• Use charts and graphs to show trends.</li> </ul> </li> <li>8. Generate Insights <ul style="list-style-type: none"> <li>• Convert findings into actionable improvements.</li> </ul> </li> <li>9. Share Findings</li> </ol>                                                                                                                                                                      | CO4 | 4B | K4 |

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|          | <ul style="list-style-type: none"> <li>• Present reports using visuals and user stories.</li> </ul> <p>10. Collaborate with Stakeholders</p> <ul style="list-style-type: none"> <li>• Work with teams to develop solutions.</li> </ul> <p><b>Benefits (5 Marks)</b></p> <ul style="list-style-type: none"> <li>• Better Decisions: Supports evidence-based design.</li> <li>• Priority Focus: Solves important issues first.</li> <li>• Team Collaboration: Improves communication among teams.</li> <li>• User-Centered Solutions: Creates designs matching user needs.</li> <li>• Continuous Improvement: Enables repeated UX enhancement.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |    |    |
| 5.       | <b>Discuss the fundamental principles of responsive design and examine the challenges associated with implementing them.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO4 | 4B | K3 |
| 5<br>Ans | <p><b>Definition (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Responsive Design makes websites adapt to different screen sizes and devices.</li> <li>• Provides a consistent and user-friendly experience on desktop, tablet, and mobile.</li> </ul> <p><b>Key Principles (5 Marks)</b></p> <ol style="list-style-type: none"> <li>1. Fluid Grid Layout <ul style="list-style-type: none"> <li>• Uses flexible percentage-based layouts.</li> <li>• Adjusts content according to screen size.</li> </ul> </li> <li>2. Flexible Images <ul style="list-style-type: none"> <li>• Images resize automatically.</li> <li>• Prevents overflow on different devices.</li> </ul> </li> <li>3. Media Queries <ul style="list-style-type: none"> <li>• Apply different styles based on screen size and orientation.</li> <li>• Improve readability and layout.</li> </ul> </li> <li>4. Mobile-First Approach <ul style="list-style-type: none"> <li>• Design starts with mobile screens.</li> <li>• Enhances features for larger devices.</li> </ul> </li> <li>5. Viewport Meta Tag <ul style="list-style-type: none"> <li>• Controls scaling and screen width adaptation.</li> </ul> </li> </ol> <p><b>Benefits (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Better User Experience: Consistent and accessible across devices.</li> <li>• SEO Improvement: Mobile-friendly sites rank better.</li> <li>• Cost Effective: One design works for multiple devices.</li> <li>• Higher Conversion: Smooth experience increases engagement.</li> </ul> <p><b>Challenges (3 Marks)</b></p> <ul style="list-style-type: none"> <li>• Performance Optimization: Compress images and optimize code.</li> <li>• Device Testing: Test across multiple screens and browsers.</li> <li>• Maintenance: Ensure continuous compatibility.</li> </ul> | CO4 | 4B | K3 |
| 6.       | <b>Solve how responsive design enhances user accessibility and satisfaction, and provide examples of successful implementations.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO4 | 4B | K4 |

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| 6<br>Ans | <p><b>Responsive Design (5 Marks)</b></p> <ul style="list-style-type: none"> <li>• Uses one flexible layout that adapts to different devices.</li> <li>• Uses fluid grids, flexible media, and CSS breakpoints.</li> <li>• Uses relative units like %, fr instead of fixed pixels.</li> <li>• Provides: <ul style="list-style-type: none"> <li>◦ Consistent experience across devices.</li> <li>◦ Lower maintenance cost.</li> <li>◦ Better SEO with a single website.</li> <li>◦ Future support for new screen sizes.</li> </ul> </li> <li>• Core idea: <i>One design, many shapes.</i></li> </ul> <p><b>Accessibility (5 Marks)</b></p> <ul style="list-style-type: none"> <li>• Improves usability for users with disabilities.</li> <li>• Provides: <ul style="list-style-type: none"> <li>◦ Adjustable text size and larger touch targets.</li> <li>◦ Content reflow without horizontal scrolling.</li> <li>◦ Screen reader and zoom support.</li> <li>◦ Consistent navigation for assistive technologies.</li> <li>◦ Support for touch, keyboard, and different orientations.</li> </ul> </li> <li>• Core idea: Responsiveness helps create accessible experiences.</li> </ul> <p><b>Examples (3 Marks)</b></p> <p><b>Responsive Design</b></p> <ul style="list-style-type: none"> <li>• News website changes from 4 columns (desktop) → 2 columns (tablet) → 1 column (mobile).</li> <li>• Menu changes into a hamburger icon on small screens.</li> </ul> <p><b>Accessibility</b></p> <ul style="list-style-type: none"> <li>• Banking apps use larger buttons to prevent wrong taps.</li> <li>• Government websites remain readable at 200% zoom.</li> </ul> | CO4 | 4B | K4 |
| 7.       | <p><b>Detail the steps involved in conducting usability tests in UX design and analyze their significance in the design process.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO4 | 4B | K3 |
| 7<br>Ans | <p><b>Definition (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• Usability testing is a user-centered evaluation method where real users test a product by completing tasks.</li> <li>• It identifies usability problems and measures ease of use, efficiency, and satisfaction.</li> </ul> <p><b>Key Steps (4 Marks)</b></p> <ol style="list-style-type: none"> <li>1. Define Objectives – Set testing goals and usability areas to evaluate.</li> <li>2. Recruit Participants – Select users representing the target audience.</li> <li>3. Create Tasks – Prepare realistic user scenarios.</li> <li>4. Prepare Environment – Set up tools, prototype, and testing space.</li> <li>5. Conduct Tests – Observe users and collect feedback.</li> <li>6. Record Findings – Note actions, problems, and user comments.</li> <li>7. Analyze Feedback – Identify patterns and prioritize issues.</li> <li>8. Redesign &amp; Improve – Apply changes based on findings.</li> </ol> <p><b>Best Practices (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Maintain moderator neutrality.</li> <li>• Use realistic scenarios.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO4 | 4B | K3 |



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|          | <ul style="list-style-type: none"> <li>• Conduct pilot testing before final tests.</li> <li>• Use remote testing when needed.</li> <li>• Perform multiple testing rounds.</li> <li>• Report findings clearly.</li> </ul> <b>Benefits (3 Marks)</b> <ul style="list-style-type: none"> <li>• User-Centered Design: Meets user needs and expectations.</li> <li>• Problem Identification: Finds usability issues early.</li> <li>• Data-Based Decisions: Supports evidence-based improvements.</li> <li>• Better Conversion: Increases engagement and retention.</li> <li>• Competitive Advantage: Builds user trust and loyalty.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |           |
| 8.       | <b>Explain the steps involved in usability testing, from planning and preparation through execution to analysis of test findings, and demonstrate their importance in the UX design process.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CO4</b> | <b>4B</b> | <b>K3</b> |
| 8<br>Ans | <b>Introduction (3 Marks)</b><br>Evaluative user research methods are used to assess existing designs, prototypes, or products. They help identify usability problems, validate design decisions, and improve user experience through feedback and analysis.<br><b>Methods (10 Marks)</b><br><b>1. Heuristic Evaluation</b> <ul style="list-style-type: none"> <li>• Experts review the interface using usability principles.</li> <li>• Identifies usability issues quickly and cost-effectively.</li> <li>• Helps improve designs before user testing.</li> </ul> <b>2. Cognitive Walkthrough</b> <ul style="list-style-type: none"> <li>• Evaluators analyze a product from a user's perspective.</li> <li>• Finds confusing steps and interaction problems.</li> <li>• Improves ease of task completion.</li> </ul> <b>3. Card Sorting</b> <ul style="list-style-type: none"> <li>• Users organize information into meaningful groups.</li> <li>• Helps create better information architecture.</li> <li>• Reveals user mental models and preferences.</li> </ul> <b>4. Tree Testing</b> <ul style="list-style-type: none"> <li>• Tests whether users can find information in a site structure.</li> <li>• Evaluates navigation and content organization.</li> <li>• Identifies areas causing user confusion.</li> </ul> <b>5. A/B Testing</b> <ul style="list-style-type: none"> <li>• Compares two design versions to measure performance.</li> <li>• Uses metrics like clicks and conversions.</li> <li>• Supports data-driven design decisions.</li> </ul> <b>6. Surveys and Questionnaires</b> <ul style="list-style-type: none"> <li>• Collect user opinions and satisfaction data.</li> <li>• Can reach a large number of users.</li> <li>• Helps understand user preferences.</li> </ul> <b>7. Eye Tracking Studies</b> <ul style="list-style-type: none"> <li>• Records where users look on an interface.</li> <li>• Helps analyze attention and visual hierarchy.</li> <li>• Improves placement of important elements.</li> </ul> | CO4        | 4B        | K3        |

### PART- C

| Q.No  | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CO's | Unit | Bloom's Level |
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| 1.    | <b>Evaluate the effectiveness of usability tests as a method for identifying design issues, and justify the extent to which they contribute to enhancing the overall user experience compared to other evaluative methods.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO4  | 4C   | K5            |
| 1 Ans | <p><b>Definition (2 Marks)</b><br/>Usability testing is a user-centered evaluation method where real users perform tasks on a product to identify usability problems and provide feedback. It measures how easy, efficient, and satisfying the product is to use.</p> <p><b>Key Steps in Conducting Usability Tests (4 Marks)</b></p> <ol style="list-style-type: none"> <li><b>Define Objectives</b> <ul style="list-style-type: none"> <li>Identify what aspects of usability need evaluation.</li> </ul> </li> <li><b>Recruit Participants</b> <ul style="list-style-type: none"> <li>Select users representing the target audience.</li> </ul> </li> <li><b>Create Tasks and Scenarios</b> <ul style="list-style-type: none"> <li>Prepare realistic tasks based on user goals.</li> </ul> </li> <li><b>Prepare Testing Environment</b> <ul style="list-style-type: none"> <li>Set up tools, prototypes, and a suitable testing space.</li> </ul> </li> <li><b>Conduct Test Sessions</b> <ul style="list-style-type: none"> <li>Observe users while they complete tasks.</li> </ul> </li> <li><b>Observe and Record</b> <ul style="list-style-type: none"> <li>Note user actions, difficulties, and feedback.</li> </ul> </li> <li><b>Collect Feedback</b> <ul style="list-style-type: none"> <li>Gather opinions and suggestions from participants.</li> </ul> </li> <li><b>Iterate and Redesign</b> <ul style="list-style-type: none"> <li>Analyze results and improve the design.</li> </ul> </li> </ol> <p><b>Best Practices (4 Marks)</b></p> <ul style="list-style-type: none"> <li><b>Moderator Neutrality:</b> Avoid influencing user behavior.</li> <li><b>Realistic Scenarios:</b> Use tasks similar to real-life situations.</li> <li><b>Pilot Testing:</b> Test the process before the actual study.</li> <li><b>Remote Testing:</b> Use online tools for flexible testing.</li> <li><b>Iterative Testing:</b> Repeat testing for continuous improvement.</li> <li><b>Clear Reporting:</b> Present findings clearly to the design team.</li> </ul> <p><b>Benefits of Usability Testing (3 Marks)</b></p> <ul style="list-style-type: none"> <li><b>User-Centered Design:</b> Creates products that satisfy user needs.</li> <li><b>Issue Identification:</b> Finds usability problems early.</li> <li><b>Data-Driven Decisions:</b> Supports decisions using real user data.</li> <li><b>Improved Conversion:</b> Increases engagement and retention.</li> <li><b>Competitive Advantage:</b> Builds customer trust and loyalty.</li> </ul> | CO4  | 4C   | K5            |
| 2.    | Design a usability testing plan for a real or hypothetical product feature, then construct a case for how the findings it would generate could be synthesized into specific product improvements,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CO4  | 4C   | K6            |

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|       | projecting their likely impact on user experience.                                                                                         |     |    |    |
| 2 Ans | <b>STUDENTS MAY EXPLAIN ANY REALTIME EXAMPLES</b><br>Testing Plan (7 Marks)<br>Findings & Improvements (4 Marks)<br>Real Example (3 Marks) | CO4 | 4C | K6 |

**UNIT- 5**  
**RESEARCH, DESIGNING, IDEATING, & INFORMATION ARCHITECTURE**  
**PART A**

| Q.No  | Questions                                                                                                                                             | CO's | Unit | Bloom's Level |
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| 1.    | <b>Why is it essential to clearly define a problem statement in the design process?</b>                                                               | CO5  | 5A   | K2            |
| 1 Ans | A well-defined problem statement provides focus, guiding the design process by identifying the specific challenges that need to be addressed.         | CO5  | 5A   | K2            |
| 2.    | <b>What role does a problem statement play in stakeholder communication?</b>                                                                          | CO5  | 5A   | K2            |
| 2 Ans | A problem statement facilitates effective communication among stakeholders by succinctly articulating the design challenge and its significance.      | CO5  | 5A   | K2            |
| 3.    | <b>Name a qualitative research method commonly used in UX design.</b>                                                                                 | CO5  | 5A   | K1            |
| 3 Ans | User interviews are a common qualitative research method in UX design.                                                                                | CO5  | 5A   | K1            |
| 4.    | <b>Define what a persona represents in UX design.</b>                                                                                                 | CO5  | 5A   | K1            |
| 4 Ans | A persona is a fictional character representing a user type, helping to humanize and understand the diverse needs of the target audience.             | CO5  | 5A   | K1            |
| 5.    | <b>How do personas contribute to user-centered design?</b>                                                                                            | CO5  | 5A   | K2            |
| 5 Ans | Personas guide design decisions by providing a detailed understanding of users' goals, behaviors, and preferences.                                    | CO5  | 5A   | K2            |
| 6.    | <b>Why is divergent thinking encouraged during solution ideation?</b>                                                                                 | CO5  | 5A   | K2            |
| 6 Ans | Divergent thinking generates a wide range of creative ideas, fostering innovation and exploration during the ideation phase.                          | CO5  | 5A   | K2            |
| 7.    | <b>How do user stories contribute to Agile development?</b>                                                                                           | CO5  | 5A   | K2            |
| 7 Ans | User stories provide concise requirements that align with user needs, facilitating Agile development by breaking down features into manageable units. | CO5  | 5A   | K2            |
| 8.    | <b>What is the primary purpose of a flow diagram in UX design?</b>                                                                                    | CO5  | 5A   | K1            |
| 8 Ans | Flow diagrams visually represent the sequential steps a user takes to complete a task or achieve a goal within a system.                              | CO5  | 5A   | K1            |

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| <b>9.</b>     | <b>Define Information Architecture in UX design.</b>                                                                                                                     | <b>CO5</b> | <b>5A</b> | <b>K1</b> |
| <b>9 Ans</b>  | Information Architecture involves organizing and structuring content within a system to enhance usability and findability for users.                                     | CO5        | 5A        | K1        |
| <b>10.</b>    | <b>Why is framing a problem statement as a challenge preferable in UX design?</b>                                                                                        | <b>CO5</b> | <b>5A</b> | <b>K2</b> |
| <b>10 Ans</b> | Framing a problem statement as a challenge encourages a solution-oriented mindset, motivating designers to explore creative solutions.                                   | CO5        | 5A        | K2        |
| <b>11.</b>    | <b>Name a usability testing method used to observe users in their natural environment.</b>                                                                               | <b>CO5</b> | <b>5A</b> | <b>K1</b> |
| <b>11 Ans</b> | Remote usability testing allows users to participate from their own locations, providing insights into real-world usage scenarios.                                       | CO5        | 5A        | K1        |
| <b>12.</b>    | <b>State the purpose of Wireflow Creation in interface design.</b>                                                                                                       | <b>CO5</b> | <b>5A</b> | <b>K1</b> |
| <b>12 Ans</b> | Wireflow Creation combines wireframes with user flow to illustrate screen navigation and interactions. It helps designers visualize the user journey before development. | CO5        | 5A        | K1        |
| <b>13.</b>    | <b>Explain the importance of Prototype Testing before product development.</b>                                                                                           | <b>CO5</b> | <b>5A</b> | <b>K2</b> |
| <b>13 Ans</b> | Prototype Testing helps identify usability issues and gather user feedback before development. This reduces design errors and improves the overall user experience.      | CO5        | 5A        | K2        |
| <b>14.</b>    | <b>Identify the objective of Heuristic Evaluation.</b>                                                                                                                   | <b>CO5</b> | <b>5A</b> | <b>K1</b> |
| <b>14 Ans</b> | The objective of Heuristic Evaluation is to assess the usability of a user interface using recognized usability principles and identify potential design issues.         | CO5        | 5A        | K1        |
| <b>15.</b>    | <b>Why is Content Strategy important in UX design?</b>                                                                                                                   | <b>CO5</b> | <b>5A</b> | <b>K2</b> |
| <b>15 Ans</b> | Content Strategy ensures that content is structured, accessible, and aligned with user needs, resulting in an improved overall user experience.                          | CO5        | 5A        | K2        |

#### PART- B

| <b>Q.No</b>  | <b>Questions</b>                                                                                                                                                                                                                                                                                                                                               | <b>CO's</b> | <b>Unit</b> | <b>Bloom's Level</b> |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|----------------------|
| <b>1.</b>    | <b>Explain the significance of a well-crafted problem statement in the UX design process. Discuss how the clarity and specificity of a problem statement influence user research, ideation, and solution development.</b>                                                                                                                                      | <b>CO5</b>  | <b>5B</b>   | <b>K3</b>            |
| <b>1 Ans</b> | <b>Definition (2 Marks)</b><br>A problem statement clearly describes the user's problem and guides the UX design process toward solving user needs.<br><b>Significance (8 Marks)</b> <ul style="list-style-type: none"> <li>• Defines design goals and challenges.</li> <li>• Helps understand user needs.</li> <li>• Aligns team and stakeholders.</li> </ul> | <b>CO5</b>  | <b>5B</b>   | <b>K3</b>            |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |    |    |
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|       | <ul style="list-style-type: none"> <li>• Supports better decisions.</li> <li>• Reduces unnecessary work and risks.</li> <li>• Guides research methods.</li> <li>• Supports idea generation.</li> <li>• Improves solution development and testing.</li> </ul> <p><b>Influence on UX Process</b></p> <ul style="list-style-type: none"> <li>• <b>User Research:</b> Identifies users and selects suitable research methods.</li> <li>• <b>Ideation:</b> Guides brainstorming of user-focused solutions.</li> <li>• <b>Solution Development:</b> Helps create and test effective designs.</li> </ul> <p><b>Example (3 Marks)</b><br/> <b>Problem:</b> Users find hospital appointment booking difficult due to complex registration.<br/> <b>Solution:</b> Research user needs, create simple booking flow, prototype, and test improvements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |    |    |
| 2.    | <p><b>Discuss the selection of appropriate UX research methods. Explain the criteria for choosing qualitative and quantitative research methods.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CO5 | 5B | K3 |
| 2 Ans | <p><b>Introduction</b><br/> UX research methods help understand user needs, behavior, and expectations. Choosing the right method helps create better design decisions.</p> <p><b>Selection Criteria (5 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Research Objective:</b> Choose methods based on study goals.</li> <li>• <b>Type of Data:</b> Qualitative for insights, quantitative for numbers.</li> <li>• <b>Project Stage:</b> Use qualitative early; quantitative for validation.</li> <li>• <b>Sample Size:</b> Small groups for qualitative, large groups for quantitative.</li> <li>• <b>Time and Budget:</b> Select methods based on available resources.</li> </ul> <p><b>Qualitative Methods (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Understand user feelings and experiences.</li> <li>• Examples: <ul style="list-style-type: none"> <li>○ Interviews</li> <li>○ Focus groups</li> <li>○ Observation</li> <li>○ Think-aloud testing</li> </ul> </li> </ul> <p><b>Example:</b> Interviewing patients about hospital booking problems.</p> <p><b>Quantitative Methods (4 Marks)</b></p> <ul style="list-style-type: none"> <li>• Collect measurable user data.</li> <li>• Examples: <ul style="list-style-type: none"> <li>○ Surveys</li> <li>○ Questionnaires</li> <li>○ A/B testing</li> <li>○ Website analytics</li> </ul> </li> </ul> | CO5 | 5B | K3 |

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| 3.    | <b>Analyze the role of personas in human-centered design. Explain the process of creating personas and discuss how they support empathy-driven design with suitable examples.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CO5 | 5B | K4 |
| 3 Ans | <b>Definition (2 Marks)</b><br>A persona is a fictional user profile created from real research data. It represents user goals, needs, behaviors, and challenges.<br><b>Role of Personas (8 Marks)</b> <ul style="list-style-type: none"> <li>• Represent target user characteristics.</li> <li>• Understand user goals and pain points.</li> <li>• Guide user-centered design decisions.</li> <li>• Reduce assumptions in design.</li> <li>• Improve team communication.</li> <li>• Ensure better usability.</li> <li>• Help create products matching user needs.</li> <li>• Improve user satisfaction.</li> </ul> <b>Creating Personas</b> <ol style="list-style-type: none"> <li>1. Conduct user research.</li> <li>2. Analyze user behavior.</li> <li>3. Group similar users.</li> <li>4. Create persona profiles.</li> <li>5. Validate and update personas.</li> </ol> <b>Example (3 Marks)</b><br><b>Persona:</b> Priya, 28, working professional, uses hospital apps for appointments.<br><b>Needs:</b> Quick booking, simple interface, reminders.<br><b>Design Solution:</b> One-click booking, easy navigation, notifications. | CO5 | 5B | K4 |
| 4.    | <b>Evaluate the importance of solution ideation in UX design. Discuss various ideation techniques such as brainstorming and mind mapping that contribute to innovative design solutions.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CO5 | 5B | K5 |
| 4 Ans | <b>Definition (3 Marks)</b><br>Solution ideation is the process of creating and refining ideas to solve user problems and develop effective UX solutions.<br><b>Importance of Solution Ideation (5 Marks)</b> <ul style="list-style-type: none"> <li>• Encourages creativity and innovation.</li> <li>• Generates multiple possible solutions.</li> <li>• Focuses on user needs.</li> <li>• Supports teamwork and collaboration.</li> <li>• Improves usability.</li> <li>• Enhances user experience.</li> </ul> <b>Ideation Techniques</b> <ol style="list-style-type: none"> <li><b>1. Brainstorming</b> <ul style="list-style-type: none"> <li>• Generates many ideas without criticism.</li> <li>• Encourages creative solutions.</li> </ul> </li> <li><b>2. Mind Mapping</b> <ul style="list-style-type: none"> <li>• Organizes ideas around a central problem.</li> <li>• Shows relationships between concepts.</li> </ul> </li> <li><b>3. Sketching</b> <ul style="list-style-type: none"> <li>• Creates quick visual ideas.</li> <li>• Helps compare interface designs.</li> </ul> </li> </ol>                                    | CO5 | 5B | K5 |

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|       | <b>Example (5 Marks)</b><br><b>Food Delivery App:</b> <ul style="list-style-type: none"> <li>• Ideas: One-click ordering, live tracking, personalized suggestions.</li> <li>• Mind maps organize ideas.</li> <li>• Sketches help create the final user-friendly design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |    |    |
| 5.    | <b>Explain the significance of user stories and scenarios in UX design. Discuss how they help understand user requirements and improve communication among stakeholders with suitable examples.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO5 | 5B | K4 |
| 5 Ans | <b>Definition (2 Marks)</b><br>User stories describe user needs, while user scenarios explain how users interact with a product in real situations.<br><b>Significance of User Stories (3 Marks)</b> <ul style="list-style-type: none"> <li>• Capture user requirements.</li> <li>• Prioritize features.</li> <li>• Support Agile development.</li> <li>• Focus on user goals.</li> </ul> <b>Example:</b><br><i>As a customer, I want to book a doctor appointment online so that I can avoid waiting.</i><br><b>Significance of User Scenarios (3 Marks)</b> <ul style="list-style-type: none"> <li>• Describe user interactions.</li> <li>• Understand user behavior and tasks.</li> <li>• Identify problems and improvements.</li> </ul> <b>Example:</b><br>Patient opens hospital app → selects doctor → books slot → receives confirmation.<br><b>Understanding User Requirements (3 Marks)</b> <ul style="list-style-type: none"> <li>• Identify user goals.</li> <li>• Understand workflows.</li> <li>• Find usability issues.</li> <li>• Create better solutions.</li> </ul> <b>Improving Communication (2 Marks)</b> <ul style="list-style-type: none"> <li>• Creates common understanding.</li> <li>• Improves teamwork.</li> <li>• Reduces misunderstandings.</li> <li>• Aligns stakeholders.</li> </ul> <b>Benefits</b> <ul style="list-style-type: none"> <li>• Supports user-centered design.</li> <li>• Improves collaboration.</li> <li>• Simplifies requirements.</li> <li>• Enhances usability and satisfaction.</li> </ul> | CO5 | 5B | K4 |
| 6.    | <b>Explain the purpose of flow diagrams and flow mapping in UX design. Discuss how they improve user navigation and help identify bottlenecks in a digital application.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO5 | 5B | K3 |
| 6 Ans | <b>Definition (3 Marks)</b><br>Flow diagrams and flow mapping are UX tools used to visualize user interactions, navigation paths, and task processes.<br><b>Purpose of Flow Diagrams (3 Marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO5 | 5B | K3 |

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|       | <ul style="list-style-type: none"> <li>Shows steps to complete a task.</li> <li>Explains system workflow.</li> <li>Simplifies complex processes.</li> <li>Ensures logical task flow.</li> </ul> <p><b>Purpose of Flow Mapping</b></p> <ul style="list-style-type: none"> <li>Shows complete user journeys.</li> <li>Displays alternative paths.</li> <li>Connects screens and touchpoints.</li> <li>Improves user experience.</li> </ul> <p><b>Improving Navigation (7 Marks)</b></p> <ul style="list-style-type: none"> <li>Organizes tasks in sequence.</li> <li>Reduces unnecessary steps.</li> <li>Creates clear navigation paths.</li> <li>Improves usability.</li> </ul> <p><b>Identifies Bottlenecks:</b></p> <ul style="list-style-type: none"> <li>Complex navigation.</li> <li>Extra screens/clicks.</li> <li>Broken flows.</li> <li>User confusion areas.</li> </ul> <p><b>Example</b><br/> <b>Shopping App Flow:</b><br/> Login → Search → Product → Cart → Payment → Confirmation</p> <p><b>Flow Mapping:</b></p> <ul style="list-style-type: none"> <li>Category search.</li> <li>Filters.</li> <li>Guest checkout.</li> </ul> <p><b>Improvement:</b> Simplify payment using one-click checkout.</p> <p><b>Benefits</b></p> <ul style="list-style-type: none"> <li>Better navigation.</li> <li>Less confusion.</li> <li>Faster task completion.</li> <li>Supports user-centered design.</li> </ul> |            |           |           |
| 7.    | <b>Discuss the principles and importance of Information Architecture. Explain how content organization and navigation improve usability and user satisfaction .</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>CO5</b> | <b>5B</b> | <b>K5</b> |
| 7 Ans | <p><b>Definition (2 Marks)</b><br/> Information Architecture is the process of organizing, structuring, and labeling content to help users easily find information in a digital product.</p> <p><b>Principles of IA (8 Marks)</b></p> <ul style="list-style-type: none"> <li><b>Organization:</b> Group content into meaningful categories.</li> <li><b>Hierarchy:</b> Arrange information by importance.</li> <li><b>Navigation:</b> Provide clear movement paths.</li> <li><b>Labeling:</b> Use simple and clear names.</li> <li><b>Findability:</b> Help users locate information quickly.</li> </ul> <p><b>Importance of IA</b></p> <ul style="list-style-type: none"> <li>Organizes content effectively.</li> <li>Improves navigation and task completion.</li> <li>Reduces confusion.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CO5</b> | <b>5B</b> | <b>K5</b> |



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|       | <ul style="list-style-type: none"> <li>• Enhances usability and accessibility.</li> <li>• Increases user satisfaction.</li> </ul> <p><b>Content Organization &amp; Navigation</b></p> <ul style="list-style-type: none"> <li>• Groups related information together.</li> <li>• Provides easy movement between pages.</li> <li>• Reduces clicks and improves efficiency.</li> </ul> <p><b>Example (3 Marks)</b><br/> <b>E-commerce Website:</b><br/> Products organized into categories like Electronics, Clothing, and Appliances.<br/> Users can:</p> <ul style="list-style-type: none"> <li>• Browse categories.</li> <li>• Apply filters.</li> <li>• Search products.</li> <li>• Complete purchases easily.</li> </ul> <p><b>Benefits</b></p> <ul style="list-style-type: none"> <li>• Better content access.</li> <li>• Faster navigation.</li> <li>• Improved decision-making.</li> <li>• Enhanced user experience.</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |     |    |    |
| 8.    | <b>Explain the concept of User Journey Mapping with a neat diagram. Discuss how it helps identify user pain points and enhances the overall user experience.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO5 | 5B | K4 |
| 8 Ans | <p><b>Definition (2 Marks)</b><br/> User Journey Mapping (UJM) is a visual representation of a user's interaction with a product or service. It shows user actions, emotions, touchpoints, and problems to improve the overall experience.</p> <p><b>Concept and Process (8 Marks)</b><br/> <b>Steps:</b></p> <ol style="list-style-type: none"> <li>1. <b>Identify User (Persona)</b> – Define the target user.</li> <li>2. <b>Set User Goal</b> – Understand what the user wants to achieve.</li> <li>3. <b>Map Actions</b> – List user activities step-by-step.</li> <li>4. <b>Identify Touchpoints</b> – Find interaction points with the system.</li> <li>5. <b>Find Pain Points</b> – Identify user difficulties and frustrations.</li> <li>6. <b>Improve Experience</b> – Create solutions to enhance usability.</li> </ol> <p><b>Identifying Pain Points</b></p> <ul style="list-style-type: none"> <li>• Confusing navigation</li> <li>• Slow loading time</li> <li>• Too many steps</li> <li>• User drop-off areas</li> <li>• Unmet user expectations</li> </ul> <p><b>Improving UX</b></p> <ul style="list-style-type: none"> <li>• Simplifies workflows</li> <li>• Reduces complexity</li> <li>• Improves accessibility</li> <li>• Increases user satisfaction</li> </ul> | CO5 | 5B | K4 |

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|  | <ul style="list-style-type: none"> <li>Supports user-centered design</li> </ul> <p><b>Example (4 Marks)</b></p> <p><b>Food Delivery App Journey:</b><br/> Open App → Search Restaurant → Select Food → Add Cart → Payment → Confirmation → Delivery</p> <p><b>Problem:</b> Users leave during payment.</p> <p><b>Solution:</b> Add one-click payment and multiple payment options.</p>  |  |  |  |
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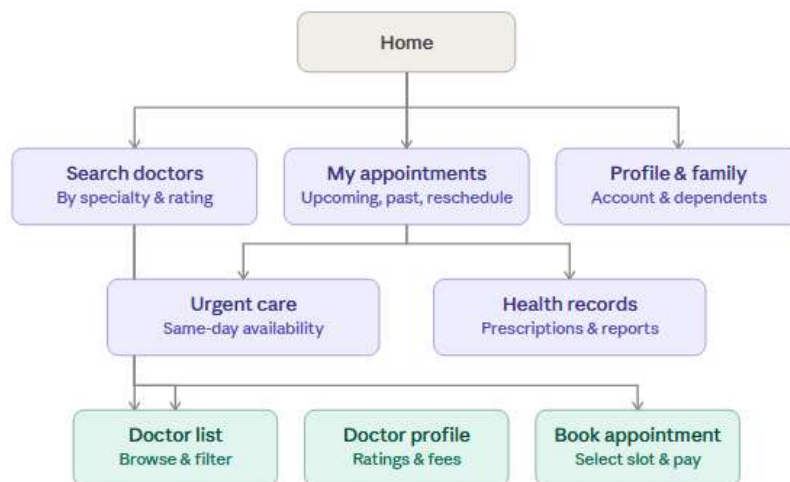
### PART- C

| Q.No | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO's | Unit | Bloom's Level |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------|
| 1(a) | Design a complete User Experience (UX) solution for an online healthcare appointment system. Formulate an appropriate problem statement, select suitable research methods, create personas, develop user stories and scenarios, prepare flow diagrams, flow mapping, Information Architecture, and a User Journey Map. Justify how each stage contributes to designing an effective user-centered solution. Illustrate your answer with suitable diagrams wherever necessary. | CO5  | 5C   | K5            |

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| 1(a) Ans | <p><b>1. Problem Statement (2 Marks)</b><br/> <b>Problem:</b><br/> Patients find appointment booking difficult due to complex processes, long waiting times, and unclear doctor information.<br/> <b>Goal:</b><br/> Create a simple system for searching doctors, booking appointments, and managing healthcare needs.</p> <p><b>2. Research Methods (2 Marks)</b></p> <ul style="list-style-type: none"> <li>• <b>Interviews:</b> Understand patient needs and problems.</li> <li>• <b>Surveys:</b> Collect feedback from many users.</li> <li>• <b>Observation:</b> Study current booking methods.</li> <li>• <b>Usability Testing:</b> Identify design issues.</li> </ul> <p><b>Contribution:</b> Helps create evidence-based, user-focused solutions.</p> <p><b>3. Persona (2 Marks)</b><br/> <b>Priya, 30, Working Professional</b><br/> <b>Goals:</b> Quick booking, doctor search, reminders.<br/> <b>Problems:</b> Long waiting, complex registration.<br/> <b>Needs:</b> Simple interface and notifications.<br/> <b>Contribution:</b> Helps designers understand user expectations.</p> <p><b>4. User Stories &amp; Scenario (2 Marks)</b><br/> <b>User Stories:</b></p> <ul style="list-style-type: none"> <li>• As a patient, I want to search doctors to find the right specialist.</li> <li>• As a patient, I want online booking to save time.</li> </ul> <p><b>Scenario:</b><br/> Open app → Search doctor → Select slot → Pay → Get confirmation.</p> <p><b>Flow Diagram: (2 Marks)</b></p> | CO5 | 5C | K5 |
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### Information Architecture: (3 Marks)



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| 2     | <p><b>Develop a user-centered UX design workflow for an e-commerce mobile application. Create wireflows and an interactive prototype, and explain how Prototype Testing, Heuristic Evaluation, and Content Strategy can be applied to evaluate and improve usability, accessibility, and overall user experience. Support your answer with suitable diagrams and examples.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO5 | 5C | K6 |
| 2 Ans | <p><b>E-commerce Mobile App UX Workflow (10 Marks)</b></p> <p><b>1. UX Workflow Stages</b></p> <ul style="list-style-type: none"> <li>• Discover → user research, competitor audit</li> <li>• Define → personas, problem statement</li> <li>• Ideate → user stories, IA, wireflows</li> <li>• Design → wireframes → interactive prototype</li> <li>• Evaluate → usability testing, heuristic evaluation</li> <li>• Iterate → refine based on findings</li> </ul> <p><b>2. Wireflow</b> (diagram above)</p> <ul style="list-style-type: none"> <li>• Combines wireframe + flow: shows <i>screens</i> and <i>navigation logic</i> together</li> <li>• Purple = browse stage, coral = transaction stage, teal = post-purchase</li> <li>• Dashed line = "edit cart" loop back from checkout</li> </ul> <p><b>3. Interactive Prototype</b> (widget above)</p> <ul style="list-style-type: none"> <li>• Clickable "Add to cart" simulates real tap behavior</li> <li>• Purpose: test flow <i>before</i> development — cheaper to fix now than in code</li> </ul> <p><b>4. Prototype Testing</b></p> <ul style="list-style-type: none"> <li>• Give users real tasks ("find and buy a size M shirt")</li> <li>• Measure: task success rate, time on task, error rate</li> <li>• Reveals: unclear labels, missed CTAs, drop-off points</li> <li>• Example: if users miss the size selector, enlarge/reposition it</li> </ul> <p><b>5. Heuristic Evaluation</b></p> <ul style="list-style-type: none"> <li>• Expert review against Nielsen's 10 heuristics (visibility of status, error prevention, consistency, etc.)</li> <li>• Fast, low-cost — no users needed</li> <li>• Example: cart icon should always show item count (visibility of system status)</li> </ul> <p><b>6. Content Strategy</b></p> <ul style="list-style-type: none"> <li>• Microcopy: clear CTAs ("Add to cart" not "Submit")</li> <li>• Error messages: say what happened + what to do next</li> <li>• Consistent terminology across screens (don't mix "Bag" and "Cart")</li> <li>• Accessibility: alt text for product images, sufficient contrast, readable font sizes.</li> </ul> <p><b>Diagram:</b></p> | CO5 | 5C | K6 |

